

THE DEFLATED SHARPE RATIO: A UNIFIED FRAMEWORK FOR SEARCH-ADJUSTED PERFORMANCE INFERENCE

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THE DEFLATED SHARPE RATIO:

A UNIFIED FRAMEWORK FOR SEARCH-ADJUSTED PERFORMANCE INFERENCE

ABSTRACT

Investment research is a search process: researchers compare factors, model specifications, machine-learning designs, parameter choices, trading rules, and managers, then focus on the strongest result. The winner of that search cannot be evaluated as though it had been specified in advance. The Deflated Sharpe Ratio (DSR) addresses this fundamental selection problem by judging reported performance against what the research process itself could have produced without skill. We unify the principal DSR implementations and explain their complementary roles. DSR-L provides the original parsimonious location benchmark for search-adjusted significance and minimum track-record analysis. DSR-LS adds the dispersion of the selected maximum, preserving a tractable location-and-scale form compatible with PSR and MinTRL while supporting power and related operating characteristics. DSR-EO uses the complete search distribution when exact tail inference is available. We also derive general calibration results linking distributional approximation error to false-positive and p-value distortion. Gaussian and Student-t benchmarks show that tail shape can alter both the magnitude and direction of approximation error, while a general GEV-domain result establishes asymptotic calibration of DSR-LS when its reference shape is matched to the extreme-value limit, with a plug-in extension under the stated scale-relative consistency conditions.

HIGHLIGHTS

- Search-adjusted inference must reflect the effective search intensity, cross-trial dispersion, and other distributional characteristics of the research process, not merely the final Sharpe ratio or the literal number of backtests.
- DSR-L, DSR-LS, and DSR-EO are complementary: a parsimonious location benchmark, a tractable location-and-scale representation compatible with PSR/MinTRL and power analysis, and exact tail inference.
- Within the finite-variance GEV region, matching the DSR-LS reference shape to the extreme-value limit yields asymptotic calibration under the stated moment-regularity condition; the result survives plug-in estimation under scale-relative consistency.
- DSR is naturally aligned with selecting one winner after search; when multiple strategies are approved or deployed, SFDR-based objectives may be more appropriate.

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JEL Classification: G0, G1, G2, G15, G24, E44.

A portfolio manager does not usually receive a research memo saying, “We tested a single strategy and obtained a Sharpe ratio of 1.5.” More often, the reported strategy is the survivor of a much larger process. A quantitative research team may have tested hundreds of signals, transformations, lookback windows, portfolio constructions, and execution rules. A machine-learning team may have varied features, architectures, hyperparameters, validation schemes, and random seeds. An allocator may have screened many managers before selecting one. Modern investment research creates repeated opportunities to observe an unusually favorable result, even when the underlying ideas have little or no skill.

The economic problem is a winner’s curse. A strategy selected because it was the best performer in a broad search is expected to look better in sample than a strategy specified before the data were examined. Two strategies can therefore have the same estimated Sharpe ratio but very different evidentiary strength. The relevant question is not only how large the reported Sharpe ratio is, but how much opportunity the research process created for an extreme result.

The Deflated Sharpe Ratio (DSR; Bailey and López de Prado 2014) was introduced to address this problem. Its central principle is straightforward: a reported Sharpe ratio should be evaluated relative to the performance that the research process itself could have generated in the absence of skill. The Probabilistic Sharpe Ratio (PSR; Bailey and López de Prado 2012) addresses uncertainty in estimating a strategy’s Sharpe ratio from a finite track record; DSR adds the separate adjustment required because the strategy was selected from alternatives.

This principle applies directly when one factor, model, strategy, or manager is promoted because it is the winner of a search. That familywise setting is the natural domain of DSR. If an organization instead approves many strategies or managers concurrently or sequentially, a false-discovery objective may be more appropriate, see López de Prado et al. (2026). The distinction is economic as well as statistical because the decision rule determines which error probability the investor should control. Exhibit 1 summarizes common investment settings in which search adjustment arises.

Exhibit 1. Where Search-Adjusted Performance Inference Arises in Practice

Investment setting	What is searched	What is ultimately reported or selected	Why conventional Sharpe inference can mislead
Factor discovery	Signals, transformations, lags, universes, portfolio rules	Best factor or factor combination	The winner reflects both skill and selection among many alternatives
Machine learning	Features, architectures, hyperparameters, loss functions, seeds	Best validation/backtest model	Model selection turns an apparently single test into a research search
Quantitative strategy development	Lookbacks, filters, risk targets, rebalancing rules, execution assumptions	Best backtest specification	Iterative refinement increases the chance of a favorable realization under the null
Hedge fund or manager due diligence	Funds, managers, share classes, track-record windows	Top-ranked manager when one winner is selected	DSR fits single-winner selection; if several managers are hired, an SFDR objective may be more natural

SOURCE: Authors.

This paper makes four contributions. First, it organizes the 2014 and 2026 DSR constructions around the common investment problem they solve: inference on a Sharpe ratio that has survived a search. The implementations are unified here as complementary representations of the same search-adjusted null distribution, each retaining a different amount of information and supporting different operational choices.

Second, the paper provides a practical map from investment objective to implementation. DSR-L is the original parsimonious location-based construction for search-adjusted significance and minimum track-record analysis. DSR-LS retains both location and scale, preserving a tractable Sharpe-ratio representation compatible with PSR and MinTRL while also supplying the dispersion needed for power and related operating characteristics. DSR-EO uses the complete finite-sample search distribution when exact upper-tail inference is available under the maintained model.

Third, the paper derives new finite-sample and asymptotic calibration results for moment-based search adjustments. Under independent Gaussian search, calibration depends on the reference shape paired with the DSR-LS location and scale: a Gaussian reference yields a 6.583% limiting rejection probability at a nominal 5% level, whereas the EVT-consistent standardized Gumbel reference is asymptotically calibrated under this benchmark. Under independent Student-t search, the Gaussian-reference rule may be conservative or liberal depending on the degrees of freedom, whereas the EVT-consistent standardized Fréchet reference is asymptotically calibrated for every fixed $\nu > 2$. More generally, Supplementary Theorem 1 establishes a GEV-domain calibration result: for independent search whose selected maximum lies in a finite-variance generalized extreme-value domain and satisfies the stated moment-regularity condition, pairing the finite-K DSR-LS location and scale with the corresponding standardized GEV reference yields asymptotically calibrated inference. Supplementary Corollary 1 shows that this conclusion survives plug-in estimation when location and scale errors are negligible relative to the selected-maximum scale and the estimated extreme-value shape is consistent. The Gaussian/Gumbel and Student-t/Fréchet results are therefore special cases of a common principle, while the finite-K comparisons show that extreme-value approximations need not be exact at modest search intensities.

Fourth, the accompanying Supplementary Material establishes several additional theoretical results. It formalizes the informational nesting of the DSR representations, showing that the location, location-and-scale, and complete-distribution constructions retain progressively more information about the same search-adjusted null, without implying a corresponding ordering of size, power, or approximation accuracy. It also proves that the calibration distortion induced by using an approximate search distribution is exactly its CDF approximation error at the induced critical value, with an analogous uniform identity for approximate p-values, and shows that finitely many moments cannot in general determine an exact upper-tail critical value. Further results clarify that a mean-zero cross-sectional skill distribution does not, in general, imply the familywise null and place DSR within the broader multiple-testing framework, including the equivalence between the exact independent-Gaussian order-statistic threshold and the Šidák correction under aligned assumptions.

HOW SEARCH CHANGES THE PERFORMANCE HURDLE

Conventional Sharpe-ratio inference is designed for a strategy treated as fixed before the data are examined: it accounts for sampling uncertainty in that strategy, but not for the bias introduced by comparing multiple alternatives and selecting one among them. Search changes the problem because the reported strategy is often the best performer among many candidates. Even when every candidate has zero true skill, the best observed Sharpe ratio will generally be positive simply because the researcher selected the maximum.

The relevant null distribution is therefore the distribution of the best result produced by the research process, not the sampling distribution of an isolated strategy. This is the common inferential target of all DSR implementations. The search penalty is determined not simply by how many models were tried, but by how much opportunity the process created for an extreme result. That opportunity depends on the effective number of trials (K), the dispersion of Sharpe ratios across trials, and the sampling law of the Sharpe-ratio estimates.¹

Operationally, every DSR implementation can be described through four elements: the inferential target, the representation retained from that target, the rule that converts the representation into a decision, and the resulting calibration and power properties. The selected-maximum formulation and the representation/decision-rule decomposition used here are stated formally in Supplementary Material Sections S1–S2; the formal DSR-L and DSR-LS constructions and their informational relationship are developed in Sections S3–S5.

A Simple Illustration

Consider a researcher evaluating strategies using five years of monthly returns ($T=60$), and suppose that every candidate has a true Sharpe ratio of zero. Under the stylized Gaussian benchmark in which each estimated monthly Sharpe ratio has standard deviation $1/\sqrt{60}$ a strategy specified before observing the data needs an estimated Sharpe ratio of about 0.212—or 0.736 when annualized—to clear a one-sided 5% significance threshold. If the researcher instead reports the best of $K=10$ trials, the corresponding 5% search-adjusted threshold rises to about 0.332, or 1.148 annualized. With $K=100$ the search-adjusted threshold rises to 0.424, or 1.468 annualized, and with $K=1,000$ trials to 0.501, or 1.737 annualized. Thus, an annualized Sharpe ratio of 1.0 may constitute strong evidence for a single strategy specified in advance, yet be entirely insufficient when that strategy is the winner of even a modest research search. The evidentiary hurdle must rise with the opportunity the research process had to generate an extreme result by chance.

The raw number of backtests should not automatically be interpreted as the number of independent experiments. Candidate strategies often share signals, data, portfolio rules, or model architecture. Conversely, a smaller search over genuinely different specifications can create

¹In empirical applications, a trial takes place whenever a researcher computes a Sharpe ratio. Trials need not be independent, and K denotes the effective number of trials: the number of independent trials that would generate the same search-induced effect as the actual, potentially dependent, set of candidate specifications. In the stylized independent-search benchmarks and the corresponding Supplementary results, however, K denotes the literal number of independent candidates; those benchmark results do not automatically extend to an effective- K approximation under dependence.

substantial dispersion across Sharpe ratios and therefore substantial opportunity for an extreme winner. The 2014 and 2026 DSR formulations emphasize both effective number of trials and cross-sectional dispersion across trials.

CHOOSING AMONG DSR IMPLEMENTATIONS

The DSR family is easiest to use when the implementation is chosen from the investment question backward. Each implementation evaluates the selected strategy against the same search-adjusted null, but it retains a different amount of information from that null and therefore supports different operational decisions. Exhibit 2 maps the three implementations to the investment decisions they are designed to support.

Exhibit 2. Choosing Among DSR Implementations

Implementation	Information retained from the search null	Primary investment use	Main advantage	Scope / trade-off
DSR-L	Expected maximum $\mu_K = E_0[M_K]$	Search-adjusted significance; minimum track-record analysis	Parsimonious benchmark shift within PSR	Retains the expected-maximum benchmark rather than the full search distribution
DSR-LS	Mean and standard deviation (μ_K, σ_K)	Search-adjusted PSR/MinTRL in location-and-scale form; power and related operating characteristics	Preserves standardized Sharpe-ratio architecture while adding search dispersion	Finite-K calibration depends on the chosen reference shape
DSR-EO	Complete finite-sample CDF F_{M_K}	Exact upper-tail critical values and p-values under the maintained search model	Uses the complete search distribution; exact under its stated model	Requires the full null distribution or a sufficiently accurate numerical representation

SOURCE: Authors.

To state the three implementations in common notation, let c index the winning specification, so that $\widehat{SR}_c$ is the realized selected maximum; under the maintained global null, let M_K denote the corresponding maximum random variable. In the order-statistic equations K is literal; effective K under dependence approximates the search effect rather than defining an exact identity.

$$M_K := \max_{1 \leq i \leq K} \widehat{SR}_i, \quad F_{M_K}(x) := P_0(M_K \leq x).$$

$$\mu_K := E_0[M_K], \quad \sigma_K := \sqrt{\text{Var}_0(M_K)}.$$

Here T is the sample size, and P_0 , E_0 , and Var_0 denote probability, expectation, and variance under the global null. Let $s_c := SE(\widehat{SR}_c)$ denote the selected strategy's PSR sampling standard error, including applicable sample-length, non-normality, and serial-dependence adjustments. By contrast, σ_K is the standard deviation of the selected maximum across repeated null searches: a search-distribution property, not the sampling standard error of one strategy. Thus DSR-L uses the search-adjusted location μ_K together with the PSR sampling scale s_c , DSR-LS uses the search-adjusted location and scale (μ_K, σ_K) , and DSR-EO uses the complete search distribution F_{M_K} . Dependence of the search distribution on T and other maintained sampling features is suppressed.

For notational consistency, each probability-valued DSR can be written as one minus its upper-tail p-value (or p-value analogue); a level- α rejection is therefore $DSR \geq 1 - \alpha$.

DSR-L: Search-Adjusted Significance and Minimum Track Record

The original location-based DSR retains the expected performance of the best strategy under the no-skill search, see Bailey and López de Prado (2014). Its practical role is to replace the conventional zero benchmark with the performance that selection alone is expected to produce. PSR continues to account for finite-sample uncertainty in the observed Sharpe ratio, while the DSR benchmark accounts for the fact that the strategy was selected from alternatives.

Using the common notation above, the location-based probability-valued statistic is

$$DSR_L(\widehat{SR}_c) = \text{PSR}(\widehat{SR}_c; SR^* = \mu_K, T) = \Phi\left(\frac{\widehat{SR}_c - \mu_K}{s_c}\right).$$

Here Φ denotes the standard Normal CDF used by the PSR sampling approximation. At nominal level α , the equivalent hurdle is

$$c_{\alpha,K}^L = \mu_K + s_c z_{1-\alpha}, \quad z_{1-\alpha} := \Phi^{-1}(1 - \alpha).$$

This makes the role of DSR-L explicit: multiplicity shifts the benchmark from zero to μ_K , while the sampling scale remains the one used by PSR for the selected strategy.

This separation answers two distinct investment questions. The sampling question asks how uncertain the estimated Sharpe ratio is because the track record is finite. The search question asks how high the best Sharpe ratio could be simply because the researcher examined multiple candidates. DSR-L is especially natural when an investment committee wants a parsimonious search-adjusted significance measure or a minimum track record length (MinTRL).

DSR-L achieves parsimony by retaining the expected maximum of the search distribution. This one-moment representation is designed to provide the original search-adjusted benchmark within the PSR framework; richer distributional information is supplied by the DSR-LS and DSR-EO representations when the investment decision calls for it.

DSR-LS: Tractable Location-and-Scale Inference

DSR-LS retains both the expected maximum and its dispersion, providing a tractable location-and-scale representation of the search-adjusted null. López de Prado et al. (2026) develop this representation to extend standardized Sharpe-ratio tools such as PSR and MinTRL to search-adjusted inference while incorporating search-induced dispersion. Retaining the first two moments preserves this compatibility without prescribing a unique reference shape. The reference distribution is therefore an operational choice within DSR-LS: a Gaussian reference preserves continuity with conventional standardized Sharpe-ratio inference; under independent Gaussian search, the standardized Gumbel provides the EVT-consistent alternative; and under independent Student-t search, the standardized Fréchet provides the corresponding EVT-consistent alternative. More generally, the Supplementary Material shows that, for independent search in a finite-variance GEV domain and under the stated moment-regularity condition, matching the standardized

reference to the extreme-value shape yields asymptotic calibration, with a corresponding plug-in result under scale-relative consistency of the estimated inputs. Their calibration properties are examined later in the paper.

Let H denote a generic continuous standardized reference CDF with mean zero and unit variance. H is not necessarily Gaussian: Φ is the special case corresponding to a Gaussian reference, while the standardized Gumbel and standardized Fréchet CDFs used later are alternative choices of H . The corresponding probability-valued location-and-scale statistic is

$$\text{DSR}_{LS,H}(\widehat{SR}_c) = H\left(\frac{\widehat{SR}_c - \mu_K}{\sigma_K}\right).$$

and its level- α hurdle is

$$c_{\alpha,K}^{LS,H} = \mu_K + \sigma_K H^{-1}(1 - \alpha).$$

Accordingly, the 2026 Gaussian operationalization is the special case $H = \Phi$. The standardized Gumbel and Fréchet references used below retain the same μ_K and σ_K while changing only the reference shape H .

The location parameter raises the performance benchmark to reflect the expected effect of search, while the scale parameter captures the variability of the selected maximum across repeated searches. Together, they permit search-adjusted PSR and MinTRL calculations to remain within a standardized Sharpe-ratio framework and provide the dispersion required for prospective power and related operating-characteristic calculations. Power additionally requires a specified alternative, because the null location and scale alone do not determine the probability of approving a genuinely skilled strategy.

DSR-LS therefore occupies the middle ground between the parsimonious DSR-L benchmark and the complete DSR-EO distribution. By retaining the first two moments of the search-adjusted null, it preserves the tractability of a location-and-scale representation and compatibility with standardized Sharpe-ratio inference, while exact finite- K tail calibration generally requires additional information about the upper tail; DSR-EO supplies this information by retaining the complete search distribution. The resulting approximation accuracy depends on the chosen reference shape and can be assessed against the exact search distribution when available, or against simulation and other richer representations when it is not. When exact finite- K tail inference is required and the complete search distribution is available, DSR-EO provides the corresponding full-distribution benchmark.

DSR-EO: Exact Tail Inference Under a Specified Search Model

DSR-EO retains the complete finite-sample null distribution of the selected maximum. Under a specified search model, it delivers the exact upper-tail critical value and the corresponding search-adjusted p-value. The independent-Gaussian order-statistic construction was derived in López de Prado, Lipton, and Zoonekynd (2026); we use the label DSR-EO here for taxonomic convenience. For continuous search distributions, DSR-EO uses the complete search distribution directly:

$$\text{DSR}_{EO}(\widehat{SR}_c) = F_{M_K}(\widehat{SR}_c), \quad p_{EO}(\widehat{SR}_c) = 1 - F_{M_K}(\widehat{SR}_c).$$

with level- α hurdle

$$c_{\alpha,K}^{EO} = F_{M_K}^{-1}(1 - \alpha).$$

Under independent search, if candidate i has null CDF F_i , the complete search distribution has the exact order-statistic form

$$F_{M_K}(x) = \prod_{i=1}^K F_i(x).$$

For i.i.d. trials with common null CDF F_0 , this reduces to $F_{M_K}(x) = F_0(x)^K$; under standardized independent-Gaussian search, $F_0 = \Phi$, so $F_{M_K}(x) = \Phi(x)^K$.

Thus, when the maintained search model correctly specifies the distribution of the selected maximum, DSR-EO is the exact probability transform of that maximum and its upper-tail p-value is exact.

For practitioners, DSR-EO is the natural benchmark when the research process is sufficiently well specified that its full null distribution can be derived or computed. In more complicated settings, the same full-distribution idea can be implemented numerically through integration, bootstrap, or Monte Carlo methods. Together, DSR-L, DSR-LS, and DSR-EO span parsimonious, tractable, and full-distribution implementations of the same search-adjusted problem.

A PRACTITIONER WORKFLOW FOR SEARCH-ADJUSTED INFERENCE

Define the Research Search

The relevant research object is the process that generated the reported result. A research log should preserve the candidate set containing the specifications that materially influenced the final choice, and the selection rule. In factor research this may include transformations, lags, universes, and portfolio rules. In machine learning it may include features, architectures, hyperparameters, validation schemes, and repeated training runs. Once the unsuccessful specifications and selection path disappear, the final Sharpe ratio alone cannot reveal how much search produced it.

This does not imply that every line of code is an independent experiment. The search model should reflect the opportunity to generate an extreme winner. That opportunity depends on the effective number of trials and the cross-sectional dispersion of Sharpe ratios across trials.

State the Investment Decision

Different investment questions require different statistical objects. If one strategy is promoted because it is the best result from a search, the natural question is familywise: how likely is a winner this strong under the no-skill search? That is the fundamental question DSR was designed to answer. Conversely, if many strategies or managers are approved or deployed, the relevant objective may instead be a sequential false discovery rate (SFDR) control, see López de Prado et al. (2026). A research head asking how likely a genuinely skilled strategy is to clear the hurdle is asking a power question. An allocator asking for the probability that a selected manager truly has

skill is asking a posterior question that also requires assumptions about the prevalence and magnitude of skill.

Choose the Representation That Matches the Decision

If the objective is a parsimonious search-adjusted significance benchmark, DSR-L may be sufficient. If the organization wants a tractable location-and-scale representation that remains compatible with PSR and MinTRL and also supplies dispersion for power analysis, DSR-LS is the natural extension. If exact tail inference is required and the full null distribution can be derived or computed, DSR-EO is the natural reference. More information is useful only when it is relevant to the decision and can be modeled credibly.

Evaluate the Complete Procedure

Location and scale define the representation retained by DSR-LS; calibration, size, and power additionally depend on the operational rule, including the chosen reference distribution. DSR-EO instead uses the complete order-statistic distribution under its maintained model.

Stress-Test the Search Model

If the research process involves dependence, non-Gaussianity, heteroskedasticity, adaptive model selection, or other departures from the analytical benchmark, the search distribution should be reassessed under assumptions that better reflect the actual process. Numerical integration, bootstrap, Monte Carlo, and richer parametric approximations can all be used within the same DSR framework. A holdout sample is not automatically a cure: if researchers repeatedly consult the holdout while refining the model, those validation decisions become part of the search and should be reflected in the inferential design.

Why Tail Accuracy Matters

A tractable approximation need not reproduce the entire search distribution; it must be sufficiently accurate where the investment decision is made, particularly in the upper tail where approval thresholds are set. Tail sensitivity should therefore be assessed against the exact search distribution when available, or against simulation and other numerical benchmarks in more complex settings. The Supplementary Material formalizes the relationship between distributional approximation error and inferential calibration.

LESSONS FROM GAUSSIAN AND STUDENT-T SEARCH BENCHMARKS

Independent Gaussian Search

The independent-Gaussian search provides a setting in which the exact distribution of the selected maximum is known, making it possible to compare exact and moment-based rules transparently. To isolate the effects of representation and distributional shape, we can study a location-only rule and two location-and-scale rules that use the same mean and standard deviation but different reference shapes: Gaussian and EVT-consistent standardized Gumbel. This analysis isolates representation and distributional shape from the additional components incorporated in the

operational 2014 and 2026 formulas, including Sharpe-ratio sampling uncertainty and applicable skewness, kurtosis, serial-dependence, and cross-strategy variance adjustments. In Exhibit 3, K denotes the number of independent candidate strategies, μ_K the expected maximum under the no-skill null, and σ_K the corresponding standard deviation. Exhibit 3 reports the finite- K rejection rates of the exact benchmark and the three stylized approximations.

Exhibit 3. Finite-K Calibration Under Independent Gaussian Search

K	μ_K	σ_K	Exact DSR-EO benchmark	Stylized location analogue	LS with Gaussian reference	LS with Gumbel reference
5	1.162964	0.668980	5.0000%	1.2408%	5.7659%	3.9129%
10	1.538753	0.586808	5.0000%	0.7249%	5.9736%	4.1446%
100	2.507594	0.429424	5.0000%	0.1643%	6.3388%	4.5776%
1,000	3.241436	0.351362	5.0000%	0.0514%	6.4709%	4.7529%

SOURCE: Authors' calculations. Entries are null rejection probabilities at a nominal 5% level.

The location-only analogue is strongly conservative under independent Gaussian search. The comparison of the two LS reference shapes shows that neither reference uniformly dominates at finite K . At $K = 5$, the LS rule with a Gaussian reference has a rejection probability of 5.7659%, closer to the exact 5% benchmark than the 3.9129% obtained with the standardized Gumbel reference. By $K = 10$, the ordering reverses slightly: the Gumbel reference yields 4.1446%, compared with 5.9736% for the Gaussian reference. The Gumbel-reference rejection probability moves progressively toward 5% at the larger K values reported in Exhibit 3. This crossover is consistent with the asymptotic nature of the extreme-value approximation: for a small search, the standardized maximum need not yet be well approximated by its Gumbel limit; as the number of independent candidates grows, it approaches a Gumbel rather than a Gaussian distribution.

Proposition 5 in Supplementary Material Section S7.7, with the asymptotic proof in Appendix B, establishes that the LS rule with a Gaussian reference approaches a rejection probability of approximately 6.583% at a nominal 5% level. That 6.583% figure is the limiting rejection profile of the Gaussian-reference stylized analogue. Holding the same DSR-LS location and scale fixed and instead using the standardized Gumbel limit yields asymptotically calibrated inference under independent Gaussian search: the rejection probability converges to the nominal level. The Gumbel approximation remains asymptotic rather than finite- K exact because convergence of Gaussian maxima to their extreme-value limit is itself asymptotic. These rejection probabilities characterize stylized reference-shape analogues that isolate the moment-and-shape component; they are not the rejection probabilities of the complete operational 2014 or 2026 DSR procedures. Thus, the location-and-scale representation and the distributional-shape choice are distinct modeling components and distinct potential sources of approximation error.

Student-t Search

In this section we extend the preceding analysis to a heavy-tailed search distribution. Under independent Student-t search with $\nu > 2$ degrees of freedom, the exact finite- K distribution of the selected maximum is available, while its large-search extreme-value limit is Fréchet rather than

Gumbel. This setting therefore permits the same comparison made under Gaussian search: hold the DSR-LS location and scale fixed, and vary only the reference distribution used to translate those moments into a critical value.

We consider two location-and-scale rules. The first pairs the mean and standard deviation of the selected maximum with a Gaussian reference, preserving the conventional standardized Sharpe-ratio form. The second pairs the same mean and standard deviation with the standardized Fréchet distribution implied by the Student-t tail index ν . The latter is the EVT-consistent reference under Student-t search, just as the standardized Gumbel is the EVT-consistent reference under Gaussian search.

The Supplementary Material shows that, for fixed $\nu > 2$, the Student-t maximum divided by its extreme-value normalization converges to a Fréchet random variable and that its first two moments converge as well. Consequently, the selected maximum standardized by its own mean and standard deviation converges to the corresponding standardized Fréchet distribution. Pairing DSR-LS with that reference therefore yields asymptotically calibrated inference: at any fixed $\nu > 2$, its rejection probability converges to the nominal level as K increases. The Gaussian-reference rule need not do so; depending on ν , its limiting rejection probability can be either below or above the nominal level.

Exhibit 4. Finite-K Calibration Under Independent Student-t Search

ν	$K = 10$	$K = 100$	$K = 1,000$	$K \rightarrow \infty$
2.5	3.1338% / 5.2377%	3.0497% / 5.0375%	3.0360% / 5.0061%	3.0334% / 5.0000%
3	4.3152% / 5.2339%	4.1936% / 5.0522%	4.1652% / 5.0114%	4.1572% / 5.0000%
4	5.3681% / 5.1574%	5.2566% / 5.0612%	5.2110% / 5.0201%	5.1889% / 5.0000%
5	5.8009% / 5.0573%	5.7396% / 5.0498%	5.6930% / 5.0226%	5.6584% / 5.0000%
10	6.2072% / 4.7043%	6.3428% / 4.9310%	6.3486% / 4.9827%	6.3068% / 5.0000%

SOURCE: Authors' calculations. Entries are null rejection probabilities at a nominal 5% level. Within each cell, the first figure uses the Gaussian reference and the second uses the standardized Fréchet reference.

The comparison sharpens the role of distributional shape within DSR-LS. Across every (ν, K) combination reported in Exhibit 4, the standardized Fréchet reference is closer to the exact 5% benchmark than the Gaussian reference. For heavier tails, the Gaussian rule is conservative, whereas for larger ν it becomes liberal; the Fréchet-reference rule instead approaches 5% systematically as search intensity increases. The finite-K Fréchet approximation is not exact, because convergence to the extreme-value limit is itself asymptotic, and the exhibit should not be interpreted as establishing uniform finite-K dominance outside the cases examined.

Together with the Gaussian results, the Student-t benchmark reveals a common structure. Under Gaussian search, the selected maximum belongs to the Gumbel domain of attraction, and the corresponding EVT-consistent Gumbel reference is asymptotically calibrated. Under Student-t search, the maximum belongs to the Fréchet domain of attraction, and the corresponding standardized Fréchet reference is asymptotically calibrated. Thus, the location-and-scale representation does not by itself determine calibration, complementing Supplementary Proposition

4: the first two moments do not determine the relevant upper tail by themselves. The general result in Supplementary Theorem 1 shows that this principle is not confined to the two parametric benchmarks: within the finite-variance GEV region, the appropriate standardized extreme-value reference yields asymptotically calibrated inference under the stated moment-regularity condition.

The Supplementary Material provides the formal finite-sample Student-t expressions and moment-convergence results, and then generalizes the Gaussian/Gumbel and Student-t/Fréchet calculations. Supplementary Theorem 1 gives the GEV-domain calibration result, Corollary 1 establishes the plug-in version under scale-relative consistency, and Corollary 2 identifies the Gaussian/Gumbel and Student-t/Fréchet cases as specializations of the same theorem.

APPLICATIONS IN INVESTMENT RESEARCH

Factor Discovery

Consider a factor research program that begins with a broad set of economic signals. Researchers examine alternative normalizations, neutralization schemes, winsorization rules, lags, holding periods, regions, and portfolio constructions. The final factor may be economically motivated, but its reported performance is still the outcome of a search.

The search-adjusted question is not whether the final factor's Sharpe ratio exceeds zero in isolation. It is whether that Sharpe ratio is unusual relative to the best result the same research process could have produced under a no-skill benchmark. The appropriate DSR representation then depends on the decision being made and on how much of the search distribution can be modeled credibly.

DSR is especially appropriate when one discovery supersedes competing alternatives or becomes a common foundation, whereas SFDR is better suited when multiple qualifying factors are deployed concurrently or sequentially.

Machine-Learning Model Selection

Machine-learning research makes the selection problem especially visible. A production model may be the endpoint of feature selection, hyperparameter optimization, architecture selection, early stopping, validation design, and repeated model training. Reporting only the final backtest can make a large adaptive search look like a single experiment. The research log should record each performance evaluation that could influence model selection.

The DSR perspective forces the research process into the inferential record. If two models have the same final Sharpe ratio but one emerged from a small pre-specified candidate set and the other from thousands of adaptive experiments, the evidentiary burden should not be identical. Because candidate models are often highly dependent, an effective-number-of-trials approximation or simulation of the actual model-selection pipeline may be more appropriate than a literal count of model runs. Repeated reuse of a validation or holdout sample should also be recorded because it can become another layer of search and selection.

Quantitative Strategy Development

A systematic trading strategy may be refined through choices about signal horizon, entry and exit rules, volatility scaling, turnover constraints, transaction-cost assumptions, portfolio construction, and execution. Each choice can improve the backtest for legitimate economic reasons while also increasing the opportunity for selection bias.

DSR does not require the researcher to pretend that development did not occur. It asks the researcher to model the consequences of development. The search adjustment should reflect the effective number of trials, cross-trial dispersion, and other distributional characteristics created during the development process, with the representation chosen according to the decision and the information available.

Hedge Fund and Manager Due Diligence

Suppose an allocator screens a large universe of managers and hires the fund with the strongest historical Sharpe ratio. If the question is whether a result this strong could have appeared simply because the allocator selected the best track record from many candidates, the DSR framework is directly relevant. The same logic applies when an investment committee chooses one strategy that supersedes the alternatives.

The decision changes if the allocator hires several managers or if an investment committee approves qualifying strategies sequentially. In that setting, a false-discovery objective may be more natural than familywise DSR. The next section distinguishes these decision settings and the additional assumptions required for posterior or false-discovery inference.

WHAT DSR CONTROLS—AND WHAT IT DOES NOT

Familywise Inference

The core DSR question is familywise: under a null in which every candidate strategy satisfies the benchmark, how unusual is it for the selected maximum to cross the approval threshold? This is DSR's native decision setting: one factor, model, strategy, or manager is promoted because it is the best performer from a collection of candidates, so the investor cares about the probability of a false winner.

DSR is naturally formulated against a no-skill null in which none of the candidate strategies has a true Sharpe ratio above the benchmark. In the standard sharp-null formulation, all candidate Sharpe ratios are zero. If one or more candidates genuinely has positive skill, the null is false and DSR should reject with increasing probability as the evidence strengthens. The purpose of the search adjustment is therefore not to assume that skill is absent in reality, but to determine whether the performance of the selected winner is too strong to be plausibly attributed to selection among unskilled candidates.

Posterior Skill and False Discovery

A posterior probability of skill requires a model for both the prevalence and the performance distribution of genuine signals. Under search, that model must operate at the trial level. For example, López de Prado and Fabozzi (2026) let each of the candidate specifications be drawn from either a null or an alternative distribution, so that the number of genuine signals among the trials is random. The familywise null corresponds to no genuine signals, while the alternative averages over cases with different numbers of genuine signals, weighted by their respective probabilities. This produces a search-adjusted posterior false-discovery probability for the reported winner. Consequently, when the search process is hidden, both the prevalence of genuine signals and the search-and-selection mechanism must be modeled—or the selection link must be broken with genuinely independent validation data.

Power and Decision Costs

A conservative research hurdle reduces false positives but can also reject genuinely profitable strategies. Power matters whenever the organization has material aversion towards missed opportunities. The appropriate trade-off is economic as well as statistical: a false deployment may be costly when implementation is expensive, capacity is scarce, or operational risk is material, while a missed strategy may be costly when the opportunity set is limited. DSR-LS and richer full-distribution procedures help quantify this trade-off under specified alternatives.

IMPLICATIONS FOR RESEARCH GOVERNANCE AND REPORTING

The unified framework suggests several changes in how quantitative research should be documented and evaluated. First, the search process should be treated as inferential data, not merely administrative metadata. A final Sharpe ratio does not reveal the unsuccessful specifications, effective number of trials, cross-trial dispersion, or selection rule that produced it. Research organizations should therefore preserve enough information about candidate generation, model refinement, validation, and selection to reconstruct a defensible search-adjusted null. This is especially important when the search is latent, because the reported winners alone cannot identify the process that generated them, or the false discovery rate (López de Prado and Fabozzi 2026).

Second, research reports should state the inferential objective, the DSR representation used, and—when a location-and-scale rule is employed—the reference distribution that operationalizes it. Tail sensitivity should be documented when consequential: the Gaussian and Student-t search benchmarks show that calibration depends on distributional shape. The Gaussian benchmark shows that the same location and scale can yield different calibration under Gaussian and EVT-consistent Gumbel references. The Student-t benchmark shows that the same location and scale can yield different calibration under Gaussian and EVT-consistent Fréchet references. When an EVT-based reference is estimated rather than fixed by the benchmark model, the assumed extreme-value domain, shape estimate, and estimation procedure should also be reported. Exact or simulation-based benchmarks can be reported when the tail assumption materially affects the decision.

Third, the error criterion should match how capital is actually allocated. DSR's familywise objective is most natural when one winner supersedes the alternatives or when the decision can have systemic consequences, whereas SFDR may better match settings in which many strategies or managers are approved or when individual decisions have localized consequences. Search-adjusted significance should not be presented as a posterior probability of skill, which requires additional assumptions about the prevalence and distribution of genuine skill.

Finally, DSR should influence the evidentiary hurdle rather than the performance measure itself. The Sharpe ratio remains a measure of risk-adjusted performance; PSR asks whether that estimate is reliable given the strategy's own track record, while DSR asks whether the evidence remains exceptional after accounting for the research process that selected the strategy. Exhibit 5 converts these governance principles into a practitioner decision guide.

Exhibit 5. Practitioner Decision Guide

Practical question	Recommended starting point	Why
Is the single best strategy significant after accounting for the research search?	DSR-L, DSR-LS, or DSR-EO	Choose among parsimonious, location-and-scale, and full-distribution representations according to the decision and information available
Do we want PSR/MinTRL in a tractable form that also reflects dispersion of the selected maximum?	DSR-LS	Retains location and scale, preserving standardized Sharpe-ratio inference while incorporating more information about the search
How likely are we to approve a genuinely skilled strategy under a specified alternative?	DSR-LS or DSR-EO	Power requires search dispersion plus an alternative model
Are multiple strategies or managers approved or deployed, without systemic consequences?	SFDR framework rather than familywise DSR alone	The economic target is the proportion or posterior probability of false approvals, not the familywise event for one winner
What is the probability that the selected manager or strategy truly has skill?	Observed posterior / false-discovery (oFDR) model plus an explicit search model	A p-value or familywise DSR does not identify posterior skill probability; hidden search must also be modeled
Are candidates strongly dependent, non-Gaussian, or adaptively selected?	Hierarchical clustering, spectral methods or simulation-based search distribution	Effective number of trials, cross-trial dispersion, and tail shape should reflect the actual research process

SOURCE: Authors.

CONCLUSION

Investment research is a selection process. Factors, models, strategies, and managers are often reported because they performed favorably relative to alternatives. Once this is recognized, the relevant benchmark changes: the selected result should be compared with what the research process could have produced in the absence of skill, not with the distribution of an isolated strategy fixed in advance. The size of that adjustment depends on the effective opportunity for an extreme result, including effective number of trials, cross-trial dispersion, and other distributional characteristics of the search.

The DSR family offers complementary ways to implement this principle. DSR-L retains the expected maximum and provides the original parsimonious location benchmark. DSR-LS also retains the dispersion of the selected maximum, preserving a tractable location-and-scale representation compatible with PSR and MinTRL while supplying inputs for power and related operating characteristics. DSR-EO retains the complete finite-sample search distribution and supplies exact upper-tail inference when that distribution is available.

The calibration results clarify the trade-off among these representations. Approximate inference is governed by how well the approximation represents the search distribution where decisions are made. The Gaussian and Student-t benchmarks demonstrate that tail shape can materially affect calibration and that the direction of the error need not be the same across reference distributions. The Supplementary Material shows that this is not merely a feature of those two parametric examples: under independent search in the finite-variance GEV region and the stated moment-regularity condition, matching the DSR-LS location and scale with the appropriate standardized extreme-value reference yields asymptotic calibration; the conclusion is preserved under plug-in estimation when the required scale-relative consistency conditions hold. Exact and approximate implementations are therefore complementary tools, not competing definitions of search-adjusted significance.

For practitioners, the most important implication is procedural. A research organization should preserve the search that produced the winning strategy, state the economic decision before choosing an error criterion, and use a null model that reflects effective number of trials, cross-trial dispersion, and other distributional characteristics of that search. DSR is naturally aligned with choosing one winner after search, or when a false positive can have systemic repercussions; when many candidates are approved or the consequences of mistakes are localized, SFDR-based objectives may be more appropriate. Search adjustment is therefore part of research governance and capital-allocation discipline, not merely a statistical correction applied after the fact. The mathematical foundations and complete proofs are provided in the accompanying Supplementary Material.

These principles extend beyond finance. The Sharpe ratio is a form of signal-to-noise ratio, and the problem studied here—evaluating the strongest observed signal after searching across many noisy candidates—arises throughout empirical science. Whenever researchers compare models, experiments, features, treatments, or signals and then report or act on the most favorable outcome, the selection process belongs in the benchmark used to evaluate it. Information about that process is easiest to preserve during research and may be impossible to reconstruct reliably from the reported winner alone. In that sense, DSR research offers an unusual opportunity for methodological spillover from finance to other disciplines. Financial research has often imported statistical methods from other fields; selection-aware inference developed around the Sharpe ratio may be one of the comparatively rare cases in which a problem originating in finance generates tools of broader scientific relevance.

REFERENCES

Bailey, David H., and Marcos López de Prado. 2012. “The Sharpe Ratio Efficient Frontier.” *Journal of Risk* 15 (2): 3–44.

Bailey, David H., and Marcos López de Prado. 2014. “The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality.” *Journal of Portfolio Management* 40 (5): 94–107.

López de Prado, Marcos, and Frank J. Fabozzi. 2026. “The False Discovery Rate in Finance: Identification Failure and Search-Adjusted Estimation.” Working paper, SSRN 6450418.

López de Prado, Marcos, Alexander Lipton, and Vincent Zoonekynd. 2026. “Sharpe Ratio Inference: A New Standard for Decision Making and Reporting.” *Journal of Portfolio Management* 52 (6): 6–50.

SUPPLEMENTARY MATERIAL

for

THE DEFLATED SHARPE RATIO: A UNIFIED FRAMEWORK FOR SEARCH-ADJUSTED PERFORMANCE INFERENCE

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Purpose of this Supplement

This Supplementary Material contains the formal mathematical development, calibration analysis, Gaussian/Gumbel and Student- t /Fréchet benchmarks, a general extreme-value calibration theorem, a plug-in calibration corollary, numerical results, and complete proofs supporting the accompanying article. It is derived from the August 4, 2026 mathematical manuscript and updated to remain aligned with the revised main article. The original Introduction, Discussion, and Conclusion are omitted because motivation, investment interpretation, applications, and practitioner guidance are provided in the main article. The core mathematical statements and proofs from the August 4 master are preserved; the later Gumbel and Fréchet reference-shape results and the general GEV-domain extension are added without renumbering Propositions 1–7.

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S1 The Search-Adjusted Inference Problem

Let $\widehat{\text{SR}}$ denote the observed Sharpe ratio of a strategy computed from a sample of T observations. Conventional inference treats the reported strategy as if it had been specified *a priori*. Under this assumption, the Probabilistic Sharpe Ratio (PSR) of Bailey and López de Prado (2012) provides a one-sided significance measure for testing

$$H_0 : \text{SR} \leq \text{SR}^* \quad \text{against} \quad H_1 : \text{SR} > \text{SR}^*.$$

Under the maintained sampling approximation, and when the boundary $\text{SR} = \text{SR}^*$ is least favourable for this one-sided null,

$$\text{PSR}(\text{SR}^*) = 1 - p(\text{SR}^*),$$

where $p(\text{SR}^*)$ is the corresponding one-sided p -value evaluated at that boundary. For $\text{SR}^* = 0$, a large PSR provides evidence of positive risk-adjusted performance for a single, pre-specified strategy, prior to any selection across competing strategies.

Search changes the object of inference. Suppose that the research process evaluates

$$\mathcal{S} = \{S_1, S_2, \dots, S_K\},$$

under the global null hypothesis

$$H_0 : \text{SR}_1 = \dots = \text{SR}_K = 0,$$

sampling and cross-sectional variation nevertheless generate positive realized Sharpe ratios. If the reported strategy is

$$S^* = \arg \max_{1 \leq i \leq K} \widehat{\text{SR}}_i,$$

its relevant null distribution is that of the selected maximum,

$$M_K := \max_{1 \leq i \leq K} \widehat{\text{SR}}_i. \tag{1}$$

Thus, a positive observed Sharpe ratio need not indicate skill: it may be the expected consequence of selecting the largest realization among K candidates.

Let

$$F_{M_K}(x) := \mathbb{P}_0(M_K \leq x) \tag{2}$$

denote the cumulative distribution function of the maximum under the global null. Search-adjusted inference evaluates whether the reported Sharpe ratio is unusually large relative to F_{M_K} , rather than relative to the sampling distribution of an isolated strategy. The research process is therefore part of the null hypothesis.

Operationally, the problem can be organized as

$$\begin{aligned} & \text{inferential target} \longrightarrow \text{statistical representation} \\ & \longrightarrow \text{critical-value or } p\text{-value construction} \longrightarrow \text{size and power properties.} \end{aligned} \tag{3}$$

The search-adjusted null distribution defines the inferential target. For comparative purposes, an implementation may be decomposed into the information retained from that distribution and the operational rule used to translate that information into inference. Bailey and López de Prado (2014) and López de Prado et al. (2026) already specify operational formulas; equation (3) separates their existing components to facilitate comparison. The resulting procedures may then be evaluated in terms of size, power, and approximation error under their maintained assumptions.

S2 A Common Architecture for DSR Implementations

The 2014 DSR establishes the central search-adjusted principle: the reported Sharpe ratio must be evaluated relative to the null distribution F_{M_K} generated by the strategy-selection process. The subsequent constructions continue this principle while retaining different amounts of information from that distribution. The following statement makes this continuity explicit.

Definition 1 (Common DSR principle). *An implementation of the Deflated Sharpe Ratio evaluates the statistical significance of an observed Sharpe ratio by reference to the null distribution of the maximum Sharpe ratio generated by the strategy selection process.*

This statement separates the common methodology from its alternative operational implementations. The methodology is determined by the search-adjusted null distribution. An implementation is determined by the information retained from that distribution and by the rule that converts this information into a critical value, p -value, or decision. Procedures based on different summaries of F_{M_K} therefore belong to the same DSR family provided that they evaluate the reported Sharpe ratio against the same search-adjusted null.

To describe this continuation formally, consider an implementation that maps an observed Sharpe ratio into a measure of search-adjusted statistical evidence.

$$\mathcal{D} = \Psi(\widehat{\text{SR}}, \mathcal{R}(F_{M_K}), T)$$

Here $\mathcal{R}(F_{M_K})$ is the representation retained from the search-adjusted null distribution, T is the sample size, and Ψ is the operational construction. The operator $\mathcal{R}(\cdot)$ may retain the first moment, the first two moments, a higher-order moment vector, a parametric approximation, or the complete distribution. The mapping Ψ then translates that representation into statistical evidence. An implementation is therefore represented by the pair

$$\mathcal{I} = (\mathcal{R}, \Psi).$$

This formulation places the existing constructions in a common architecture. Retaining the expected maximum Sharpe ratio yields the location-based implementation. Retaining both its expectation and variance yields the location-and-scale implementation DSR-LS. Retaining the complete finite-sample distribution yields the exact order-statistic implementation DSR-EO. Higher-order moments, parametric approximations, or numerical representations can be incorporated without changing the underlying inferential target.

The choice of implementation depends on the inferential objective. The location summary supplies the search-adjusted benchmark used by the 2014 PSR-based significance and minimum-track-record-length constructions. Under their maintained probabilistic models, power and false-discovery calculations additionally require information concerning dispersion, the specified alternative, and, for posterior measures, the prevalence and distribution of genuine skill. The implementations are therefore complementary: they retain different amounts of information from the same search-adjusted null distribution.

Sections S3 and S4 develop the location-based and location-and-scale implementations. Section S5 establishes their mathematical relationship, while Section S7 compares them with the finite- K order-statistic construction DSR-EO.

S3 A Location-Based Implementation

Let M_K and F_{M_K} be as defined in equations (1) and (2). The location-based implementation retains only the first moment of the search-adjusted null distribution:

$$\mu_K := \mathbb{E}_0[M_K] = \mathcal{R}_L(F_{M_K}) \quad (4)$$

The quantity μ_K is the expected Sharpe ratio of the best-performing strategy under the global null. It measures the performance that selection alone produces on average and therefore replaces zero as the benchmark for the reported strategy.

The operational construction applies the Probabilistic Sharpe Ratio to this benchmark:

$$\text{DSR}_L = \text{PSR}(\widehat{\text{SR}}; \text{SR}^* = \mu_K).$$

where the subscript L indicates that only location is retained. In the notation of Section S2, the implementation is represented by

$$(\mathcal{R}_L, \Psi_L), \quad \mathcal{R}_L(F_{M_K}) = \mu_K, \quad \Psi_L(\widehat{\text{SR}}, \mu_K, T) = \text{PSR}(\widehat{\text{SR}}; \text{SR}^* = \mu_K).$$

Thus, the PSR continues to account for the sampling uncertainty of $\widehat{\text{SR}}$, while μ_K adjusts the benchmark for selection among K candidates. The two corrections address distinct sources of uncertainty: finite-sample estimation and strategy search.

This construction is parsimonious because it requires only one summary of F_{M_K} . It also nests conventional Sharpe-ratio inference under a centered null model. If the strategy is specified in advance, then $K = 1$. Under a centered zero-skill null satisfying $\mathbb{E}_0[\widehat{\text{SR}}_1] = 0$, $\mu_1 = \mathbb{E}_0[M_1] = 0$. Consequently,

$$\text{DSR}_L = \text{PSR}, \quad K = 1,$$

so the location-based DSR reduces to the PSR when no search has occurred.

The cost of this parsimony is that μ_K does not determine the dispersion or upper-tail shape of F_{M_K} . The location-based construction therefore evaluates the selected Sharpe ratio relative to the average outcome of the search process, with calibration depending on the maintained sampling model and the PSR decision rule. Inferential objectives that depend explicitly on the variability or shape of the selected maximum require a richer representation.

S3.1 Minimum Track Record Length

The location-based DSR can be inverted to determine the minimum sample length required for an estimated Sharpe ratio to attain significance. For significance level α , treat μ_K as the fixed search-adjusted benchmark supplied by the completed research process and define

$$\text{MinTRL}_{\alpha, K} := \min \left\{ T \in \mathbb{N} : \text{DSR}_L(\widehat{\text{SR}}, T, \mu_K) \geq 1 - \alpha \right\}.$$

Equivalently,

$$\text{MinTRL}_{\alpha, K} = \min \left\{ T \in \mathbb{N} : \text{PSR}(\widehat{\text{SR}}, T, \text{SR}^* = \mu_K) \geq 1 - \alpha \right\}.$$

MinTRL is therefore the track-record length at which the estimated Sharpe ratio becomes significant relative to the search-adjusted benchmark μ_K , under the maintained

PSR sampling model. Its computation inherits any skewness and kurtosis adjustments used for the sampling distribution of the Sharpe-ratio estimator. If a maintained model instead makes the search benchmark itself a function of T , that dependence must be carried through the minimization rather than holding μ_K fixed.

MinTRL is not a prospective power calculation. It solves for the sample length required by a specified Sharpe ratio and significance criterion, whereas power is the probability of rejection across repeated samples under a specified alternative.

S4 A Location-and-Scale Implementation

The location-based implementation retains only the expectation μ_K of the search-adjusted maximum. DSR-LS enriches this representation by also retaining its dispersion. Assume throughout this section that $\mathbb{E}_0[M_K^2] < \infty$, and let

$$\sigma_K^2 := \text{Var}_0(M_K) \quad (5)$$

denote the variance of the maximum Sharpe ratio under the global null. Together, the pair

$$\mathcal{R}_{LS}(F_{M_K}) := (\mu_K, \sigma_K) \quad (6)$$

provides a location-and-scale representation of F_{M_K} . It preserves both the expected outcome of selection and its variation across repeated realizations of the search process. López de Prado et al. (2026) operationalize this location-and-scale representation with a tractable Gaussian reference having the same first two moments as the maximum Sharpe ratio. In the unified framework used here, the retained representation and the reference shape are separated explicitly: the representation is (μ_K, σ_K) , whereas the reference distribution is part of the operational construction.

For comparative purposes, let H be a continuous standardized reference CDF with mean zero and unit variance, and write

$$q_{1-\alpha}^H := H^{-1}(1 - \alpha). \quad (7)$$

A location-and-scale critical value based on the retained moments and reference shape H is then

$$c_{\alpha, K}^{LS, H} := \mu_K + \sigma_K q_{1-\alpha}^H. \quad (8)$$

The Gaussian operationalization of López de Prado et al. (2026) corresponds to $H = \Phi$, so that $q_{1-\alpha}^H = z_{1-\alpha}$. The Gaussian and Student- t benchmarks below also consider EVT-consistent standardized Gumbel and Fréchet reference shapes while holding the same location and scale fixed.

The operational procedure can be written as

$$\text{DSR}_{LS} = \Psi_{LS}(\widehat{\text{SR}}, \mu_K, \sigma_K, T; H),$$

where Ψ_{LS} denotes the associated critical-value, p -value, or decision-rule construction and H records the reference shape when such a shape is used. Accordingly, DSR-LS is characterized by the retained representation together with its operational construction, not by the moment representation alone. This distinction separates the information retained from the null distribution from the rule used to convert that information into inference.

Relative to the location-based implementation, DSR-LS allows the dispersion of the selected maximum to enter explicitly. Under any specified location-and-scale reference,

(μ_K, σ_K) supplies null-distribution inputs for critical values and related operating characteristics; the resulting calibration additionally depends on the chosen reference shape. Power also depends on the distribution under the alternative, while false-discovery measures may depend on the prevalence and distribution of genuine skill. The first two moments therefore support these calculations but do not determine them without further probabilistic assumptions.

The relationship between the two representations is

$$F_{M_K} \longrightarrow \mathcal{R}_{LS}(F_{M_K}) = (\mu_K, \sigma_K).$$

Both implementations refer to the same search-adjusted null distribution; they differ in the information retained and in the operational rule applied to it. The resulting hierarchy is:

- **Location representation:** $\mathcal{R}_L(F_{M_K}) = \mu_K$; see equation (4).
- **Location-and-scale representation:** $\mathcal{R}_{LS}(F_{M_K}) = (\mu_K, \sigma_K)$; see equation (6).
- **Higher-order moment representation:** may additionally retain the skewness γ_K , kurtosis κ_K , or further distributional summaries.
- **Exact representation:** retains the complete finite-sample distribution F_{M_K} .

Thus, DSR-LS does not replace the location-based construction. It is a richer, still tractable implementation, while DSR-EO provides the exact finite- K benchmark. Section S5 establishes the mathematical relationship between the location and location-and-scale constructions.

S5 Mathematical Relationship between the Implementations

Sections S3 and S4 represent the same search-adjusted null distribution at different levels of resolution. The complete distribution F_{M_K} contains all probabilistic information generated by the selection process, while the location and location-and-scale implementations retain only selected features of it. Their relationship is therefore one of informational nesting:

$$\mathcal{R}_L(F_{M_K}) \preceq_{\text{info}} \mathcal{R}_{LS}(F_{M_K}) \preceq_{\text{info}} \cdots \preceq_{\text{info}} F_{M_K}, \quad (9)$$

where $\preceq_{\text{info}}$ denotes informational nesting: the representation on the left can be recovered from the representation on the right. Each successive representation retains additional information about the same search-adjusted null distribution.

This hierarchy separates the inferential target from its representation. DSR-EO retains the complete finite- K distribution and therefore provides the natural benchmark for evaluating moment-based approximations such as DSR-LS.

Proposition 1. *Every DSR implementation can be classified according to the statistical representation it employs for the common search-adjusted null distribution.*

The proof is given in Appendix A.

Proposition 2. *The location representation is informationally nested within the corresponding location-and-scale representation.*

The proof is given in Appendix A.

Informational nesting does not, by itself, imply an ordering of test size, power, or approximation accuracy. Those properties depend on the operational rule, the sampling model, and any distributional approximation applied to the retained representation. They must therefore be established for the complete procedure rather than inferred from the number of moments retained.

The framework also permits richer representations. For example, one may retain higher-order moments,

$$\mathcal{R}_H(F_{M_K}) = (\mu_K, \sigma_K, \gamma_K, \kappa_K),$$

where γ_K and κ_K denote the skewness and kurtosis of the maximum Sharpe ratio, respectively. Alternatively, inference may use the complete finite-sample distribution F_{M_K} from equation (2). The representations form the sequence

$$\mu_K \longrightarrow (\mu_K, \sigma_K) \longrightarrow (\mu_K, \sigma_K, \gamma_K, \kappa_K) \longrightarrow F_{M_K}.$$

Each stage retains additional information about the same null distribution. A complete DSR implementation pairs the chosen representation with the rule that translates it into a critical value, p -value, or decision. Section S6 turns from this representational hierarchy to the inferential objectives served by the resulting procedures.

S6 Inferential Objectives

The appropriate DSR implementation depends on the inferential objective and on the complete procedure used to achieve it. A richer representation is not automatically preferable: it must supply the information required by the operational rule, and the resulting procedure must be evaluated under its maintained assumptions.

The location-based implementation addresses significance and minimum track record length. It asks whether the reported Sharpe ratio exceeds the expected outcome of the search process and how much data are required to satisfy a chosen significance criterion. The benchmark is μ_K , while the PSR converts that benchmark and the sampling model of the estimated Sharpe ratio into statistical evidence. Its calibration therefore depends jointly on the benchmark, the sampling model, and the decision rule. Because μ_K alone does not identify the upper tail of F_{M_K} , the procedure should not be interpreted as an exact finite- K test without additional assumptions.

Other objectives require more information. Statistical power is the probability of rejection under a specified alternative, whereas false-discovery measures also depend on the prevalence and distribution of genuine skill. DSR-LS retains (μ_K, σ_K) as null-distribution inputs for these calculations. A specified reference shape and decision rule complete the null-side operational construction, while the alternative model supplies the additional information required for power. The additional scale information supports the operational analysis; it does not redefine the search-adjusted null or, by itself, determine power or false-discovery rates.

The informational ordering in equation (9) therefore has an operational interpretation. Any quantity depending only on μ_K can be computed from either the location or location-and-scale representation, whereas a calculation requiring σ_K cannot generally be recovered from μ_K alone. This nesting does not imply an ordering of calibration, size, or power, because those properties belong to the complete procedure. DSR-EO provides the finite- K

reference against which the moment-based DSR-LS approximation can be assessed when the full order-statistic distribution is available.

The same principle extends to higher-order or exact representations. More detailed information about F_{M_K} may be required for upper-tail critical values, prediction, or other operating characteristics. The choice among implementations is therefore governed by four elements: the inferential objective, the representation retained from the search-adjusted null, the operational construction, and the maintained assumptions. Across implementations, the defining methodological feature remains unchanged: inference is conducted relative to the null distribution generated by the strategy-selection process.

S7 Calibration, Validity, and Inferential Targets

The preceding sections distinguish the search-adjusted inferential target, the representation retained from its null distribution, and the operational rule applied to that representation. This section turns to the properties of the resulting procedure. Exact calibration, conservative validity, power, familywise inference, selective inference, and posterior error control are distinct and must be evaluated against the target for which the procedure was constructed. The analysis first links calibration distortion to CDF approximation error, then explains why finitely many moments do not generally identify exact critical values, and finally applies these results to exact and stylized moment-based rules under independent Gaussian and Student- t search.

S7.1 Exact Calibration, Conservative Validity, and Operational Construction

Let $\{\mathbb{P}_\theta : \theta \in \Theta\}$ be a statistical model, let $\Theta_0 \subseteq \Theta$ denote the null parameter space, and let φ_α be a possibly randomized test indexed by nominal significance level α . Its rejection-probability function is $\pi_\alpha(\theta) := \mathbb{E}_\theta[\varphi_\alpha]$. At $\theta \in \Theta_0$, the test is pointwise exact, conservative, or liberal according as

$$\pi_\alpha(\theta) = \alpha, \quad \pi_\alpha(\theta) < \alpha, \quad \text{or} \quad \pi_\alpha(\theta) > \alpha.$$

The size of the test is

$$\text{size}(\varphi_\alpha) := \sup_{\theta \in \Theta_0} \pi_\alpha(\theta),$$

and level- α validity requires $\text{size}(\varphi_\alpha) \leq \alpha$. Thus, equality is not required: rejection probabilities below α indicate conservatism, not invalidity. Under a composite null, exactness may hold only at a least-favourable boundary.

Under a simple continuous null, an exact p -value satisfies

$$\mathbb{P}_\theta(p \leq u) = u, \quad 0 \leq u \leq 1.$$

Over a composite null, a valid p -value satisfies

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(p \leq u) \leq u, \quad 0 \leq u \leq 1.$$

Exact uniformity is therefore stronger than Type I error control. Define the pointwise calibration distortion by

$$\Delta_\alpha(\theta) := \pi_\alpha(\theta) - \alpha.$$

so that zero, negative, and positive values correspond to exact, conservative, and liberal behavior.

To connect this distortion to approximation error, consider a simple continuous null under which the selected statistic M_K has continuous and strictly increasing CDF F_{M_K} . Let G_K be a continuous and strictly increasing CDF used as an operational approximation to F_{M_K} . Define the corresponding approximate critical value by

$$c_{\alpha,K}^G := G_K^{-1}(1 - \alpha). \quad (10)$$

Proposition 3 (Calibration distortion and CDF approximation error). *Consider the rule that rejects when $M_K > c_{\alpha,K}^G$, and define its calibration distortion by*

$$\Delta_{\alpha,K}^G := \mathbb{P}_0(M_K > c_{\alpha,K}^G) - \alpha.$$

Then

$$\Delta_{\alpha,K}^G = G_K(c_{\alpha,K}^G) - F_{M_K}(c_{\alpha,K}^G). \quad (11)$$

Moreover, if

$$p_K^G := 1 - G_K(M_K),$$

then

$$\sup_{0 < u < 1} \left| \mathbb{P}_0(p_K^G \leq u) - u \right| = \sup_{x \in \mathbb{R}} |F_{M_K}(x) - G_K(x)|. \quad (12)$$

The proof is given in Appendix B. Equation (11) shows that the pointwise calibration distortion of an approximate rule is exactly its CDF approximation error evaluated at the critical value selected by that rule. Equation (12) gives the corresponding uniform statement for approximate upper-tail p -values.

Calibration is a property of the complete operational procedure, not of the retained representation alone. This distinction does not add a previously absent layer to DSR: the earlier DSR papers already pair their representations with critical-value, probability, or decision-rule constructions. The present decomposition makes those existing components explicit so that their statistical properties can be discussed separately. For example, a location-and-scale representation may be converted into the reference-shape critical value

$$c_{\alpha}^H = \mu + \sigma H^{-1}(1 - \alpha),$$

with the Gaussian choice $H = \Phi$ giving $c_{\alpha}^H = \mu + \sigma z_{1-\alpha}$. Its rejection probability depends on the true null shape, the chosen reference shape, and the maintained assumptions. The same principle applies to the location and DSR-LS representations: size and power arise only after the critical-value, p -value, or decision rule has been specified.

S7.2 Moments and Exact Critical Values

At nominal level α , the exact upper-tail critical value is

$$c_{\alpha,K}^{\text{ex}} := F_{M_K}^{-1}(1 - \alpha). \quad (13)$$

The corresponding exact upper-tail p -value is

$$p_K^{\text{ex}} := 1 - F_{M_K}(M_K). \quad (14)$$

Under a simple continuous null, p_K^{ex} is uniformly distributed on $(0, 1)$. By Proposition 3 and the strict monotonicity of F_{M_K} , the approximate rule is conservative, exactly calibrated, or liberal according as

$$c_{\alpha,K}^G > c_{\alpha,K}^{\text{ex}}, \quad c_{\alpha,K}^G = c_{\alpha,K}^{\text{ex}}, \quad c_{\alpha,K}^G < c_{\alpha,K}^{\text{ex}},$$

respectively. Moments may support an approximation to this quantile, but do not generally identify it without an additional shape assumption.

Proposition 4 (Moments do not determine exact critical values). *Fix $\alpha \in (0, 1)$ and $r \in \{1, 2, \dots\}$. Let $\mathcal{F}$ contain two distributions F_1 and F_2 whose CDFs are continuous and strictly increasing and whose first r moments are finite and identical. Assume that their $(1 - \alpha)$ -quantiles differ. Then no critical-value rule depending only on those r moments can have rejection probability exactly equal to α under both distributions.*

The proof is given in Appendix B.

The proposition does not preclude useful, valid, or conservative moment-based procedures. Nor does it correct a claim made in the earlier DSR papers. López de Prado et al. (2026) present the moment-based approximation and the exact finite- K order-statistic solution separately. Proposition 4 formalizes why these two constructions need not coincide: finitely many moments do not determine an exact upper-tail quantile over a sufficiently broad distribution class.

S7.3 The Familywise Null

Let $\text{SR}_1, \dots, \text{SR}_K$ denote the population Sharpe ratios of the K candidate strategies. For benchmark SR_0 , the familywise null is

$$H_0^{\text{FW}} : \text{SR}_i \leq \text{SR}_0 \quad \text{for every } i = 1, \dots, K. \quad (15)$$

In the zero-skill case,

$$\text{SR}_0 = 0. \quad (16)$$

Let

$$\Theta_0^{\text{FW}} := \left\{ \mathbf{s} = (s_1, \dots, s_K) \in \mathbb{R}^K : s_i \leq \text{SR}_0 \text{ for every } i \right\}.$$

The relevant familywise error criterion is

$$\sup_{\mathbf{s} \in \Theta_0^{\text{FW}}} \mathbb{P}_{\mathbf{s}}(\text{at least one rejection}) \leq \alpha.$$

Under models for which the boundary $s_1 = \dots = s_K = \text{SR}_0$ is least favourable, calibration at that boundary is sufficient for level- α control over the composite familywise null. When only the best-performing strategy is reported, the familywise event occurs when the selected maximum crosses the rejection threshold while every candidate satisfies the null. The DSR is constructed for this familywise problem; a restriction on the cross-sectional average Sharpe ratio is not equivalent to equation (15).

S7.4 Mean-Zero Skill Populations Are Not Familywise Nulls

Suppose instead that true Sharpe ratios are heterogeneous and symmetric around zero, for example,

$$\text{SR}_i \sim \mathcal{N}(0, \sigma_s^2), \quad \sigma_s^2 > 0.$$

Although $\mathbb{E}[\text{SR}_i] = 0$, symmetry and independence imply

$$\mathbb{P}(\text{SR}_i \leq 0) = \frac{1}{2}, \quad \text{and} \quad \mathbb{P}(\text{SR}_i \leq 0 \text{ for all } i) = \left(\frac{1}{2}\right)^K = 2^{-K}.$$

Hence the familywise null is false in almost every replication when K is large. Rejection frequencies from such a mixture combine true and false rejections and are not Type I error rates for equations (15)–(16).

S7.5 Familywise, Selective, and Posterior Inference

Three questions must be separated. Familywise inference asks whether the selected strategy exceeds the benchmark when all candidates satisfy the null; its reference distribution is that of the maximum under the familywise null. Selective inference conditions on the winner-selection event and generally requires a different sampling distribution. Posterior inference asks whether the selected strategy has genuine skill and requires a model for the prevalence and magnitude of skill.

These targets are not interchangeable. Familywise control does not determine a posterior skill probability, and posterior evidence does not establish familywise Type I error control. Conditional, posterior, and sequential measures such as pFDR, oFDR, and SFDR therefore extend rather than redefine the familywise DSR problem.

S7.6 Familywise Error and False Discovery Measures

Let V be the number of false rejections and R the total number of rejections. Then

$$\text{FWER} = \mathbb{P}(V \geq 1), \quad \text{FDP} = \frac{V}{\max(R, 1)}, \quad \text{and} \quad \text{FDR} = \mathbb{E}[\text{FDP}].$$

FWER controls the probability of at least one false discovery, whereas FDR averages the proportion of false discoveries among rejections. A procedure designed for the selected maximum need not control posterior or sequential error across repeated approvals, and an FDR procedure need not protect against even one false deployment. The criteria may share probabilistic ingredients, but they answer different questions.

S7.7 Previously Derived Exact Benchmark and Stylized Moment-Based Analogues

The exact independent-Gaussian order-statistic distribution and its FWER-controlling threshold were derived in López de Prado et al. (2026, equations 26–27). For taxonomic convenience, the present article labels that previously derived construction DSR-EO. We use it as a transparent benchmark against which to evaluate stylized location and location-and-scale analogues. Suppose

$$\widehat{\text{SR}}_i = Z_i, \quad Z_1, \dots, Z_K \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad (17)$$

and therefore

$$M_K = \max_{1 \leq i \leq K} Z_i. \quad (18)$$

Then

$$F_{M_K}(x) = \mathbb{P}_0(M_K \leq x) = \Phi(x)^K, \quad (19)$$

where Φ and ϕ are the standard Normal distribution function and density. Define

$$z_{1-\alpha} := \Phi^{-1}(1 - \alpha). \quad (20)$$

The exact DSR-EO threshold and search-adjusted p -value are

$$c_{\alpha,K}^{\text{ex}} := \Phi^{-1}\left((1 - \alpha)^{1/K}\right). \quad (21)$$

$$p_K^{\text{ex}} := 1 - F_{M_K}(M_K) = 1 - \Phi(M_K)^K. \quad (22)$$

The latter is uniform under the simple continuous boundary null. The first two moments of the maximum are

$$\mu_K = \mathbb{E}_0[M_K] = \int_{-\infty}^{\infty} x K \phi(x) \Phi(x)^{K-1} dx, \quad (23)$$

$$\mathbb{E}_0[M_K^2] = \int_{-\infty}^{\infty} x^2 K \phi(x) \Phi(x)^{K-1} dx, \quad \sigma_K^2 = \mathbb{E}_0[M_K^2] - \mu_K^2. \quad (24)$$

To isolate the consequences of location adjustment, moment compression, and distributional-shape approximation, compare the previously derived exact DSR-EO rule with stylized analytical analogues. These analogues suppress additional components of the complete operational DSR formulas—including Sharpe-ratio sampling uncertainty and, where applicable, skewness, kurtosis, serial-dependence, and cross-strategy variance adjustments—and are not asserted to coincide algebraically with the full 2014 or 2026 procedures.

The location-only and Gaussian location-and-scale critical values are

$$c_{\alpha,K}^L := \mu_K + z_{1-\alpha}, \quad (25)$$

$$c_{\alpha,K}^{LS} := \mu_K + \sigma_K z_{1-\alpha}. \quad (26)$$

Their corresponding tests, together with the exact benchmark, are

$$\varphi_{\alpha,K}^{\text{ex}} := \mathbf{1}\{M_K > c_{\alpha,K}^{\text{ex}}\}, \quad \varphi_{\alpha,K}^L := \mathbf{1}\{M_K > c_{\alpha,K}^L\}, \quad \varphi_{\alpha,K}^{LS} := \mathbf{1}\{M_K > c_{\alpha,K}^{LS}\}.$$

Their null rejection probabilities are

$$\begin{aligned} \mathbb{P}_0(\varphi_{\alpha,K}^{\text{ex}} = 1) &= \alpha, \\ \mathbb{P}_0(\varphi_{\alpha,K}^L = 1) &= 1 - \Phi(\mu_K + z_{1-\alpha})^K, \\ \mathbb{P}_0(\varphi_{\alpha,K}^{LS} = 1) &= 1 - \Phi(\mu_K + \sigma_K z_{1-\alpha})^K. \end{aligned}$$

The stylized location analogue is strongly conservative in this benchmark, while the stylized Gaussian location-and-scale analogue is substantially nearer to nominal size but remains mildly liberal. The exact DSR-EO construction is exact under the maintained independent-Gaussian model. The residual gap of the Gaussian analogue reflects the difference between the exact order-statistic shape and its moment-matched Gaussian approximation.

The Gumbel limit for centered and scaled Gaussian maxima is classical and is already invoked in López de Prado et al. (2026). The contribution of the following proposition is the explicit rejection-probability calculation for the stylized location and Gaussian location-and-scale analogues, not the derivation of the exact maximum distribution or its extreme-value limit.

Proposition 5 (Calibration profile of the stylized moment-based analogues). *For every fixed $\alpha \in (0, 1/2)$,*

$$\mathbb{P}_0(M_K > \mu_K + z_{1-\alpha}) = 1 - \Phi(\mu_K + z_{1-\alpha})^K \longrightarrow 0. \quad (27)$$

Let G be a standard Gumbel random variable,

$$\mathbb{P}(G \leq x) = \exp(-e^{-x}), \quad (28)$$

and let γ denote Euler's constant. Then

$$\frac{M_K - \mu_K}{\sigma_K} \xrightarrow{\mathcal{D}} \frac{G - \gamma}{\pi/\sqrt{6}}. \quad (29)$$

Consequently,

$$\mathbb{P}_0(M_K > \mu_K + \sigma_K z_{1-\alpha}) \longrightarrow 1 - \exp\left[-\exp\left\{-\gamma - \frac{\pi}{\sqrt{6}} z_{1-\alpha}\right\}\right]. \quad (30)$$

At $\alpha = 0.05$, the limiting rejection probability is approximately 0.0658301.

The standardized $(1 - \alpha)$ -quantile of the limiting Gumbel law in (29) is

$$q_{1-\alpha}^G = \frac{\sqrt{6}}{\pi} [-\log\{-\log(1 - \alpha)\} - \gamma]. \quad (31)$$

Accordingly, define the Gumbel location-and-scale analogue by

$$c_{\alpha,K}^{GLS} := \mu_K + \sigma_K q_{1-\alpha}^G. \quad (32)$$

Corollary (Asymptotic calibration of the Gumbel location-and-scale analogue). *For every fixed $\alpha \in (0, 1/2)$,*

$$\mathbb{P}_0(M_K > \mu_K + \sigma_K q_{1-\alpha}^G) \longrightarrow \alpha. \quad (33)$$

Proof. By Proposition 5,

$$\frac{M_K - \mu_K}{\sigma_K} \xrightarrow{\mathcal{D}} \frac{G - \gamma}{\pi/\sqrt{6}}.$$

The right-hand side has a continuous distribution, and $q_{1-\alpha}^G$ in (31) is its $(1 - \alpha)$ -quantile. Therefore the upper-tail rejection probability converges to α . $\square$

At $\alpha = 0.05$, $q_{0.95}^G \approx 1.86580$. The finite- K null rejection probability of the Gumbel location-and-scale analogue is

$$1 - \Phi\left(\mu_K + \sigma_K q_{1-\alpha}^G\right)^K.$$

Exact numerical integration yields the following supplementary exhibit.

Supplementary Exhibit 1: Finite- K Calibration of the Previously Derived Exact Benchmark and Three Stylized Analogues

K	μ_K	σ_K	Exact DSR-EO benchmark	Stylized location analogue	LS with Gaussian reference	LS with Gumbel reference
5	1.162964	0.668980	5.0000%	1.2408%	5.7659%	3.9129%
10	1.538753	0.586808	5.0000%	0.7249%	5.9736%	4.1446%
100	2.507594	0.429424	5.0000%	0.1643%	6.3388%	4.5776%
1,000	3.241436	0.351362	5.0000%	0.0514%	6.4709%	4.7529%

The finite- K comparison shows that neither standardized reference shape uniformly dominates. At $K = 5$, the Gaussian-reference LS rule has rejection probability 5.7659%, closer to the exact 5% benchmark than the 3.9129% obtained with the standardized Gumbel reference. By $K = 10$, the ordering reverses slightly: the Gumbel reference yields 4.1446%, compared with 5.9736% for the Gaussian reference. The Gumbel-reference rejection probability then moves progressively toward 5% at the larger K values reported in Supplementary Exhibit 1. This crossover is consistent with the asymptotic nature of the extreme-value approximation: for a small search, the standardized maximum need not yet be well approximated by its Gumbel limit.

The corollary nevertheless shows that the Gumbel-reference rule is asymptotically correctly sized under independent Gaussian search. In this benchmark, the Gaussian-reference location-and-scale analogue has asymptotic rejection probability 6.583% at nominal level 5%, corresponding to a liberal calibration gap of approximately 1.583 percentage points. This gap comes from combining the retained first two moments with a Gaussian shape assumption, not from retaining only (μ_K, σ_K) .

This conclusion does not contradict Proposition 4. That proposition states that finitely many moments do not, by themselves, determine an exact upper-tail critical value over a sufficiently broad class of distributions. The Gumbel rule adds distributional-shape information beyond the first two moments: it uses the EVT-consistent standardized Gumbel limit to determine the upper-tail quantile.

The technical appendix supplies the extreme-value, moment-convergence, and Mills-ratio arguments. The Gaussian and Gumbel comparisons separate moment compression from distributional-shape approximation: the same location and scale can yield different calibration when paired with different standardized reference shapes.

This benchmark isolates a moment-matching and shape-approximation mechanism relevant to the interpretation of DSR-LS. It does not establish that the complete operational DSR-LS procedure has exactly the rejection probabilities reported for any of the stylized analogues. Direct identification would require an explicit mapping that also accounts for Sharpe-ratio sampling uncertainty, cross-strategy search dispersion, the PSR scale, and the applicable skewness, kurtosis, and serial-dependence corrections.

DSR-EO remains the exact finite- K calibration reference already derived in López de Prado et al. (2026), while DSR-LS provides the corresponding tractable location-and-scale approximation and inputs for power and related operating-characteristic calculations. Under dependence, non-Gaussianity, or heteroskedasticity, numerical integration, bootstrap,

simulation, or richer representations may be required, and calibration must be reassessed under the corresponding model.

S7.8 Independent Student- t Search Benchmark

The preceding Gaussian benchmark can be extended by changing the marginal search distribution while retaining independence. This provides a transparent non-Gaussian benchmark for the same distinction between the retained location-and-scale representation and the reference shape used to operationalize it. Let $K \in \mathbb{N}$ and $\nu > 0$. Consider

$$X_1, \dots, X_K \stackrel{\text{iid}}{\sim} t_\nu, \quad M_{K,\nu} := \max_{1 \leq i \leq K} X_i, \quad (34)$$

and let T_ν and t_ν denote the cumulative distribution function and density of the Student- t_ν law, respectively. Independence gives

$$F_{K,\nu}(x) := \mathbb{P}(M_{K,\nu} \leq x) = T_\nu(x)^K. \quad (35)$$

Consequently, the exact level- α critical value is

$$c_{\alpha,K,\nu}^{\text{exact}} = T_\nu^{-1}\left((1-\alpha)^{1/K}\right). \quad (36)$$

For $\nu > 2$, define

$$\mu_{K,\nu} := \mathbb{E}[M_{K,\nu}] = \int_{-\infty}^{\infty} x K t_\nu(x) T_\nu(x)^{K-1} dx, \quad (37)$$

$$\sigma_{K,\nu}^2 := \text{Var}(M_{K,\nu}) = \int_{-\infty}^{\infty} x^2 K t_\nu(x) T_\nu(x)^{K-1} dx - \mu_{K,\nu}^2. \quad (38)$$

The Gaussian-reference location-and-scale rule uses

$$c_{\alpha,K,\nu}^{LS} = \mu_{K,\nu} + \sigma_{K,\nu} z_{1-\alpha}, \quad (39)$$

where $z_{1-\alpha} = \Phi^{-1}(1-\alpha)$. As in the Gaussian benchmark, this is a stylized analytical analogue designed to isolate the moment-and-shape component; it is not asserted to reproduce every component of the operational DSR-LS formula.

Proposition 6 (Finite- K Gaussian-reference Student- t calibration). *Assume $\nu > 2$ and $\alpha \in (0, 1)$. The exact null rejection probability of the Gaussian-reference location-and-scale rule is*

$$\pi_{\alpha,K,\nu}^{LS} = 1 - T_\nu(\mu_{K,\nu} + \sigma_{K,\nu} z_{1-\alpha})^K. \quad (40)$$

Moreover,

$$\pi_{\alpha,K,\nu}^{LS} \leq \alpha \iff \mu_{K,\nu} + \sigma_{K,\nu} z_{1-\alpha} \geq T_\nu^{-1}\left((1-\alpha)^{1/K}\right). \quad (41)$$

Thus, the rule is conservative, exact, or liberal according as $c_{\alpha,K,\nu}^{LS}$ is larger than, equal to, or smaller than $c_{\alpha,K,\nu}^{\text{exact}}$.

The proof is given in Appendix C. This finite- K result is also a direct specialization of Proposition 3.

For $K \geq 3$, let

$$a_{K,\nu} := T_\nu^{-1}\left(1 - \frac{1}{K}\right). \quad (42)$$

Proposition 7 (Fréchet limit and limiting Gaussian-reference calibration). *Let $\nu > 2$ and $\alpha \in (0, 1/2)$. Then*

$$\frac{M_{K,\nu}}{a_{K,\nu}} \xrightarrow{d} Y_\nu, \quad \mathbb{P}(Y_\nu \leq y) = \exp(-y^{-\nu}), \quad y > 0. \quad (43)$$

Moreover,

$$\frac{\mu_{K,\nu}}{a_{K,\nu}} \longrightarrow m_\nu := \Gamma\left(1 - \frac{1}{\nu}\right), \quad (44)$$

$$\frac{\sigma_{K,\nu}}{a_{K,\nu}} \longrightarrow s_\nu := \left[\Gamma\left(1 - \frac{2}{\nu}\right) - \Gamma\left(1 - \frac{1}{\nu}\right)^2 \right]^{1/2}. \quad (45)$$

Therefore,

$$\pi_{\alpha,K,\nu}^{LS} \longrightarrow \pi_{\alpha,\nu}^{LS,\infty} := 1 - \exp\left\{-(m_\nu + s_\nu z_{1-\alpha})^{-\nu}\right\}. \quad (46)$$

The proof is given in Appendix C. The limiting Gaussian-reference rule is non-liberal if and only if

$$\Gamma\left(1 - \frac{1}{\nu}\right) + z_{1-\alpha} \sqrt{\Gamma\left(1 - \frac{2}{\nu}\right) - \Gamma\left(1 - \frac{1}{\nu}\right)^2} \geq [-\log(1 - \alpha)]^{-1/\nu}. \quad (47)$$

The EVT-consistent alternative uses the standardized Fréchet reference implied by the Student- t tail index. The $(1 - \alpha)$ -quantile of Y_ν is

$$y_{1-\alpha}^{F_\nu} := [-\log(1 - \alpha)]^{-1/\nu}. \quad (48)$$

Hence the corresponding standardized Fréchet quantile is

$$q_{1-\alpha}^{F_\nu} := \frac{y_{1-\alpha}^{F_\nu} - m_\nu}{s_\nu} = \frac{[-\log(1 - \alpha)]^{-1/\nu} - m_\nu}{s_\nu}. \quad (49)$$

Holding the finite- K location and scale fixed, define the Fréchet-reference location-and-scale critical value by

$$c_{\alpha,K,\nu}^{FLS} := \mu_{K,\nu} + \sigma_{K,\nu} q_{1-\alpha}^{F_\nu}. \quad (50)$$

Corollary (Finite- K expression and asymptotic calibration of the Fréchet-reference rule). *Let $\nu > 2$ and $\alpha \in (0, 1/2)$. Its finite- K null rejection probability is*

$$\pi_{\alpha,K,\nu}^{FLS} := 1 - T_\nu\left(\mu_{K,\nu} + \sigma_{K,\nu} q_{1-\alpha}^{F_\nu}\right)^K, \quad (51)$$

and

$$\pi_{\alpha,K,\nu}^{FLS} \longrightarrow \alpha. \quad (52)$$

The proof is given in Appendix C. The corollary is the Student- t analogue of the Gumbel-reference result in Section S7: the location-and-scale compression is held fixed, while the standardized reference shape is matched to the appropriate extreme-value limit.

At nominal level $\alpha = 0.05$, the Gaussian-reference and standardized-Fréchet-reference rejection probabilities are reported together below. Within each cell, the first figure uses the Gaussian reference and the second uses the standardized Fréchet reference.

Supplementary Exhibit 2: Finite- K Calibration Under Independent Student- t Search at Nominal Level $\alpha = 0.05$

ν	$K = 10$	$K = 100$	$K = 1,000$	$K \rightarrow \infty$
2.5	3.1338% / 5.2377%	3.0497% / 5.0375%	3.0360% / 5.0061%	3.0334% / 5.0000%
3	4.3152% / 5.2339%	4.1936% / 5.0522%	4.1652% / 5.0114%	4.1572% / 5.0000%
4	5.3681% / 5.1574%	5.2566% / 5.0612%	5.2110% / 5.0201%	5.1889% / 5.0000%
5	5.8009% / 5.0573%	5.7396% / 5.0498%	5.6930% / 5.0226%	5.6584% / 5.0000%
10	6.2072% / 4.7043%	6.3428% / 4.9310%	6.3486% / 4.9827%	6.3068% / 5.0000%

Across every (ν, K) combination displayed in Supplementary Exhibit 2, the standardized Fréchet reference is closer to the exact 5% benchmark than the Gaussian reference. For heavier tails, the Gaussian-reference rule is conservative, whereas for larger ν it becomes liberal; the Fréchet-reference rule approaches 5% systematically as search intensity increases. The finite- K Fréchet approximation is not exact because convergence to the extreme-value limit is itself asymptotic, and the supplementary exhibit should not be interpreted as establishing uniform finite- K dominance outside the cases examined.

For the Gaussian-reference rule, numerical root finding at the five-percent level gives the limiting crossing $\nu_\infty^* \approx 3.73871$. On the examined range $2.001 \leq \nu \leq 20$, smaller degrees of freedom produce conservative behavior, while moderate tail thickness can bring the Gaussian-reference critical value closer to the exact Student- t quantile. The observed crossing and sign classification are numerical; no analytical monotonicity or uniqueness result in ν is asserted. As ν increases, the Gaussian-reference limiting profile approaches the standardized-Gumbel Gaussian benchmark.

Together, the Gaussian and Student- t benchmarks reveal a common structure. Under independent Gaussian search, the selected maximum lies in the Gumbel domain of attraction, and the EVT-consistent standardized Gumbel reference restores asymptotic calibration. Under independent Student- t search, the maximum lies in the Fréchet domain of attraction, and the corresponding standardized Fréchet reference restores asymptotic calibration. This complements Proposition 4: the first two moments do not determine the relevant upper tail by themselves, but in these benchmarks pairing the retained location and scale with the appropriate extreme-value reference supplies the additional shape information required for asymptotic calibration. The next subsection formalizes this common structure for the finite-variance region of the generalized extreme-value family and gives a plug-in result under scale-relative consistency of the estimated inputs.

S7.9 General Extreme-Value Calibration Principle

The Gaussian–Gumbel and Student- t –Fréchet calculations are instances of a more general extreme-value principle. The key distinction is between the location-and-scale information retained by DSR-LS and the limiting shape used to convert that information into a critical value.

Let $X_1, X_2, \dots$ be independent and identically distributed under the maintained null, with distribution function F , and define

$$M_K := \max_{1 \leq j \leq K} X_j. \quad (53)$$

Throughout this subsection, K is the literal number of independent candidates in the benchmark, not an effective number of dependent trials. Unless stated otherwise, all limits

are taken as $K \rightarrow \infty$, with the null distribution F fixed and with α fixed where applicable. Whenever the corresponding moments are finite, write

$$\mu_K := \mathbb{E}[M_K], \quad \sigma_K^2 := \text{Var}(M_K). \quad (54)$$

For $\xi \in \mathbb{R}$, let G_ξ denote the generalized extreme-value (GEV) distribution with zero location and unit scale. For $\xi \neq 0$,

$$G_\xi(x) = \exp\left\{-(1 + \xi x)^{-1/\xi}\right\}, \quad 1 + \xi x > 0, \quad (55)$$

with $G_\xi(x) = 0$ for $\xi > 0$ and $x \leq -1/\xi$, and $G_\xi(x) = 1$ for $\xi < 0$ and $x \geq -1/\xi$. For $\xi = 0$, define

$$G_0(x) = \exp\{-e^{-x}\}, \quad x \in \mathbb{R}. \quad (56)$$

Definition 2 (Maximum domain of attraction). *We write*

$$F \in \mathcal{D}(G_\xi)$$

if there exist sequences $a_K > 0$ and $b_K \in \mathbb{R}$ such that

$$F^K(a_K x + b_K) \longrightarrow G_\xi(x) \quad (57)$$

for every x at which G_ξ is continuous. Equivalently, if Y_ξ has distribution function G_ξ and

$$Z_K := \frac{M_K - b_K}{a_K}, \quad (58)$$

then

$$Z_K \xrightarrow{d} Y_\xi. \quad (59)$$

Weak convergence of the normalized maximum does not by itself imply convergence of its first two moments. We therefore impose an explicit moment-regularity condition. Assume that there exists $K_0 < \infty$ such that

$$\lim_{A \rightarrow \infty} \sup_{K \geq K_0} \mathbb{E}\left[Z_K^2 \mathbf{1}_{\{|Z_K| > A\}}\right] = 0. \quad (60)$$

Thus $\{Z_K^2 : K \geq K_0\}$ is uniformly integrable. The condition is asymptotic: it permits the first two moments of M_K to fail to exist for a finite number of small values of K . In the Student- t benchmark above, the required moment regularity is established directly by the Potter-bound argument in the proof of Proposition 7.

Finite-variance region. The explicit restriction $\xi < 1/2$ below makes transparent the region in which the GEV limit has a finite variance and hence admits a DSR-LS standardization. Given (59) and (60), this restriction is logically redundant: uniform integrability of Z_K^2 forces $\mathbb{E}[Y_\xi^2] < \infty$, which for a GEV limit is equivalent to $\xi < 1/2$. We retain the restriction in the theorem statement because it identifies the DSR-LS-relevant finite-variance EVT region directly.

Theorem 1 (Standardized extreme-value calibration). *Suppose that $F \in \mathcal{D}(G_\xi)$ for some $\xi < 1/2$, with norming sequences $a_K > 0$ and b_K satisfying (59), and suppose that (60) holds. Define*

$$m_\xi := \mathbb{E}[Y_\xi], \quad s_\xi := \sqrt{\text{Var}(Y_\xi)}. \quad (61)$$

Then $0 < s_\xi < \infty$, and for every $K \geq K_0$ the quantities μ_K and σ_K are finite with $\sigma_K > 0$. Moreover,

$$\frac{\mu_K - b_K}{a_K} \longrightarrow m_\xi, \quad (62)$$

$$\frac{\sigma_K}{a_K} \longrightarrow s_\xi. \quad (63)$$

Consequently,

$$\frac{M_K - \mu_K}{\sigma_K} \xrightarrow{d} \frac{Y_\xi - m_\xi}{s_\xi}. \quad (64)$$

For $\alpha \in (0, 1)$, let $q_{\xi, 1-\alpha}^*$ denote the $(1 - \alpha)$ -quantile of

$$Y_\xi^* := \frac{Y_\xi - m_\xi}{s_\xi}. \quad (65)$$

Then

$$c_{\alpha, K}^{\text{EVT}} := \mu_K + \sigma_K q_{\xi, 1-\alpha}^* \quad (66)$$

is asymptotically calibrated:

$$\mathbb{P}(M_K > c_{\alpha, K}^{\text{EVT}}) \longrightarrow \alpha. \quad (67)$$

Proof. By Definition 2, $Z_K \xrightarrow{d} Y_\xi$. By the continuous-mapping theorem, $Z_K^2 \xrightarrow{d} Y_\xi^2$. The uniform-integrability condition (60) therefore yields

$$\mathbb{E}[Z_K^2] \longrightarrow \mathbb{E}[Y_\xi^2]. \quad (68)$$

Moreover, uniform integrability of $\{Z_K^2\}$ implies uniform integrability of $\{|Z_K|\}$, and hence

$$\mathbb{E}[Z_K] \longrightarrow \mathbb{E}[Y_\xi] = m_\xi. \quad (69)$$

It follows that

$$\text{Var}(Z_K) \longrightarrow \text{Var}(Y_\xi) = s_\xi^2. \quad (70)$$

Since

$$\mathbb{E}[Z_K] = \frac{\mu_K - b_K}{a_K}, \quad \text{Var}(Z_K) = \frac{\sigma_K^2}{a_K^2},$$

we obtain (62) and (63). The restriction $\xi < 1/2$ guarantees that the GEV limit has finite variance; nondegeneracy of G_ξ implies $s_\xi > 0$. Since membership in the maximum domain of attraction of a nondegenerate GEV law excludes a degenerate marginal distribution F , M_K is nondegenerate for every finite K ; hence $\sigma_K > 0$ whenever its second moment is finite.

Next,

$$\frac{M_K - \mu_K}{\sigma_K} = \frac{Z_K - (\mu_K - b_K)/a_K}{\sigma_K/a_K}. \quad (71)$$

Together with

$$Z_K \xrightarrow{d} Y_\xi, \quad \frac{\mu_K - b_K}{a_K} \rightarrow m_\xi, \quad \frac{\sigma_K}{a_K} \rightarrow s_\xi > 0,$$

Slutsky's theorem gives

$$\frac{M_K - \mu_K}{\sigma_K} \xrightarrow{d} \frac{Y_\xi - m_\xi}{s_\xi}.$$

Finally, the standardized GEV distribution is continuous. Therefore, for every $\alpha \in (0, 1)$,

$$\begin{aligned}\mathbb{P}(M_K > \mu_K + \sigma_K q_{\xi, 1-\alpha}^*) &= \mathbb{P}\left(\frac{M_K - \mu_K}{\sigma_K} > q_{\xi, 1-\alpha}^*\right) \\ &\longrightarrow \mathbb{P}\left(\frac{Y_\xi - m_\xi}{s_\xi} > q_{\xi, 1-\alpha}^*\right) \\ &= \alpha.\end{aligned}$$

□

The GEV moment formulas used below follow directly from an exponential representation. If $E \sim \text{Exp}(1)$ and $\xi \neq 0$, then

$$Y_\xi \stackrel{d}{=} \frac{E^{-\xi} - 1}{\xi}, \quad (72)$$

and

$$\mathbb{E}[E^{-r\xi}] = \Gamma(1 - r\xi), \quad r\xi < 1. \quad (73)$$

Taking $r = 1, 2$ gives the following expressions. For $\xi \neq 0$ and $\xi < 1/2$,

$$m_\xi = \frac{\Gamma(1 - \xi) - 1}{\xi}, \quad (74)$$

$$s_\xi^2 = \frac{\Gamma(1 - 2\xi) - \Gamma(1 - \xi)^2}{\xi^2}. \quad (75)$$

At $\xi = 0$ these expressions extend continuously to

$$m_0 = \gamma, \quad s_0^2 = \frac{\pi^2}{6}, \quad (76)$$

where γ is the Euler–Mascheroni constant.

For $p \in (0, 1)$,

$$y_{\xi, p} := G_\xi^{-1}(p) = \begin{cases} \frac{[-\log p]^{-\xi} - 1}{\xi}, & \xi \neq 0, \\ -\log[-\log p], & \xi = 0. \end{cases} \quad (77)$$

Hence

$$q_{\xi, 1-\alpha}^* = \frac{y_{\xi, 1-\alpha} - m_\xi}{s_\xi}, \quad (78)$$

and

$$c_{\alpha, K}^{\text{EVT}} = \mu_K + \sigma_K \frac{y_{\xi, 1-\alpha} - m_\xi}{s_\xi}. \quad (79)$$

Corollary 1 (Plug-in calibration under scale-relative consistency). *Suppose the assumptions of Theorem 1 hold for some fixed $\xi < 1/2$. Let $\hat{\mu}_K$, $\hat{\sigma}_K$, and $\hat{\xi}_K$ be finite real-valued estimators, possibly constructed from data statistically dependent on M_K . Assume that their errors are asymptotically negligible at the DSR-LS scale in the sense that*

$$\frac{\hat{\mu}_K - \mu_K}{\sigma_K} \xrightarrow{p} 0, \quad (80)$$

$$\frac{\hat{\sigma}_K}{\sigma_K} \xrightarrow{p} 1, \quad (81)$$

$$\hat{\xi}_K \xrightarrow{p} \xi. \quad (82)$$

Because $\sigma_K > 0$ and $\hat{\sigma}_K/\sigma_K \xrightarrow{p} 1$,

$$\mathbb{P}(\hat{\sigma}_K > 0) \longrightarrow 1. \quad (83)$$

Because $\xi < 1/2$, condition (82) implies

$$\mathbb{P}(\hat{\xi}_K < 1/2) \longrightarrow 1. \quad (84)$$

To keep the finite- K rule well defined even when an estimated shape falls outside the finite-variance GEV region, define

$$\hat{q}_{\alpha,K}^{\text{EVT}} := \begin{cases} q_{\hat{\xi}_K, 1-\alpha}^*, & \hat{\xi}_K < 1/2, \\ q_{0, 1-\alpha}^*, & \hat{\xi}_K \geq 1/2, \end{cases} \quad (85)$$

where the second branch is an arbitrary fixed finite fallback and has asymptotically negligible probability by (84). Define the plug-in critical value

$$\hat{c}_{\alpha,K}^{\text{EVT}} := \hat{\mu}_K + \hat{\sigma}_K \hat{q}_{\alpha,K}^{\text{EVT}}. \quad (86)$$

Then, for every $\alpha \in (0, 1)$,

$$\mathbb{P}(M_K > \hat{c}_{\alpha,K}^{\text{EVT}}) \longrightarrow \alpha. \quad (87)$$

Proof. For fixed $\alpha \in (0, 1)$, the map

$$\eta \longmapsto q_{\eta, 1-\alpha}^*$$

is continuous on $(-\infty, 1/2)$. For $\eta \neq 0$, this follows from (74), (75), and (77), together with continuity of the gamma function and positivity of s_η . At $\eta = 0$, the same formulas have the continuous extensions in (76) and the $\eta = 0$ branch of (77). Therefore, by (82), the continuous-mapping theorem, and (84), the piecewise quantity in (85) satisfies

$$\hat{q}_{\alpha,K}^{\text{EVT}} \xrightarrow{p} q_{\xi, 1-\alpha}^*. \quad (88)$$

Normalize the plug-in threshold by the population DSR-LS scale:

$$\frac{\hat{c}_{\alpha,K}^{\text{EVT}} - \mu_K}{\sigma_K} = \frac{\hat{\mu}_K - \mu_K}{\sigma_K} + \frac{\hat{\sigma}_K}{\sigma_K} \hat{q}_{\alpha,K}^{\text{EVT}}. \quad (89)$$

Equations (80), (81), and (88) imply

$$\frac{\hat{c}_{\alpha,K}^{\text{EVT}} - \mu_K}{\sigma_K} \xrightarrow{p} q_{\xi, 1-\alpha}^*. \quad (90)$$

Theorem 1 gives

$$\frac{M_K - \mu_K}{\sigma_K} \xrightarrow{d} Y_\xi^*. \quad (91)$$

Since the normalized random threshold in (90) converges in probability to a constant, Slutsky's theorem yields

$$\frac{M_K - \mu_K}{\sigma_K} - \frac{\hat{c}_{\alpha,K}^{\text{EVT}} - \mu_K}{\sigma_K} \xrightarrow{d} Y_\xi^* - q_{\xi, 1-\alpha}^*. \quad (92)$$

The distribution of Y_ξ^* is continuous, so zero is a continuity point of the limiting distribution in (92). Hence

$$\begin{aligned}\mathbb{P}(M_K > \hat{c}_{\alpha,K}^{\text{EVT}}) &= \mathbb{P}\left(\frac{M_K - \mu_K}{\sigma_K} - \frac{\hat{c}_{\alpha,K}^{\text{EVT}} - \mu_K}{\sigma_K} > 0\right) \\ &\longrightarrow \mathbb{P}(Y_\xi^* > q_{\xi,1-\alpha}^*) \\ &= \alpha.\end{aligned}$$

No independence between $(\hat{\mu}_K, \hat{\sigma}_K, \hat{\xi}_K)$ and M_K is required; convergence in probability of the normalized threshold to a constant is sufficient. $\square$

Scope of the plug-in result. Corollary 1 does not prescribe a particular estimator of ξ , μ_K , or σ_K , and it does not establish (80)–(82) for any specific estimation scheme. Because μ_K and σ_K are K -dependent targets, the first two conditions are scale-relative rather than ordinary fixed-parameter consistency statements. In particular,

$$\hat{\mu}_K - \mu_K = o_p(\sigma_K), \quad \hat{\sigma}_K - \sigma_K = o_p(\sigma_K).$$

Thus, when $\sigma_K \rightarrow 0$, as in some Gumbel-domain examples, the required location accuracy is stronger than absolute $o_p(1)$ consistency; when σ_K diverges, absolute consistency of $\hat{\mu}_K$ is not required. These conditions must therefore be verified under the sampling, simulation, and dependence structure of the empirical implementation. The corollary states only that, once the stated scale-relative conditions and $\hat{\xi}_K \xrightarrow{p} \xi$ hold, plug-in estimation does not alter the asymptotic calibration conclusion.

Corollary 2 (Gumbel and Fréchet special cases). *Under the assumptions of Theorem 1:*

1. If $\xi = 0$, then G_ξ is the Gumbel distribution and

$$q_{0,1-\alpha}^* = \frac{\sqrt{6}}{\pi} [-\log\{-\log(1-\alpha)\} - \gamma], \quad (93)$$

which reproduces the standardized-Gumbel location-and-scale rule.

2. If F lies in a Fréchet maximum domain of attraction with extreme-value index $\xi = 1/\nu$ for some $\nu > 2$, let V_ν satisfy

$$\mathbb{P}(V_\nu \leq v) = \exp(-v^{-\nu}), \quad v > 0.$$

Then

$$Y_{1/\nu} \stackrel{d}{=} \nu(V_\nu - 1). \quad (94)$$

Consequently,

$$\frac{Y_{1/\nu} - \mathbb{E}[Y_{1/\nu}]}{\sqrt{\text{Var}(Y_{1/\nu})}} \stackrel{d}{=} \frac{V_\nu - \mathbb{E}[V_\nu]}{\sqrt{\text{Var}(V_\nu)}}, \quad (95)$$

so the standardized-Fréchet rule used in the Student- t_ν benchmark is exactly the $\xi = 1/\nu$ specialization of the theorem.

Proof. The Gumbel statement follows directly from (76) and (77).

For the Fréchet case, set $\xi = 1/\nu$ and $W := \nu(V_\nu - 1)$. For $w > -\nu$,

$$\begin{aligned}\mathbb{P}(W \leq w) &= \mathbb{P}\left(V_\nu \leq 1 + \frac{w}{\nu}\right) \\ &= \exp\left[-\left(1 + \frac{w}{\nu}\right)^{-\nu}\right] \\ &= \exp\left[-(1 + \xi w)^{-1/\xi}\right] \\ &= G_\xi(w).\end{aligned}$$

Thus $W \stackrel{d}{=} Y_{1/\nu}$. Since the transformation from V_ν to W has positive slope ν , centering and scaling by the corresponding standard deviations gives (95). $\square$

Bounded upper tails. Theorem 1 also covers the Weibull-type case $\xi < 0$, provided the moment-regularity condition (60) holds. No separate parametric family is required for the limiting reference shape.

Interpretation. Theorem 1 provides an asymptotic positive counterpart to Proposition 4. The location and scale (μ_K, σ_K) alone do not determine an exact upper-tail critical value. Once these location-and-scale inputs are available, and under the stated moment-regularity condition, asymptotic calibration does not require specification of the complete parametric family F for the reference shape: the additional limiting reference-shape information is summarized by the extreme-value domain and its shape parameter ξ . The Gaussian ($\xi = 0$) and Student- t_ν ($\xi = 1/\nu$) calculations are therefore special cases of a common extreme-value calibration principle.

S7.10 Generic Multiple-Testing Procedures

The DSR belongs to the broader family of multiple-testing corrections. Bonferroni and Šidák adjust marginal tests to control a familywise event; the DSR targets the null distribution of the selected maximum, either directly or through a retained representation of that distribution. Under independent Gaussian statistics, the Šidák threshold and equation (21) are equivalent descriptions of the same event.

Comparisons are meaningful only when the null hypothesis, dependence structure, marginal distributions, and familywise event are aligned. Otherwise, differences in rejection frequencies may reflect different probabilistic models rather than superior calibration. Once the target is common, the substantive comparison concerns the information retained from the null distribution and the rule used to convert it into inference.

S7.11 Power, Type II Error, and Decision Costs

For $\theta_1 \in \Theta_1$, the power and Type II error probabilities are

$$\pi(\theta_1) = \mathbb{E}_{\theta_1}[\varphi_\alpha].$$

$$\beta(\theta_1) = 1 - \pi(\theta_1).$$

Power depends on the specified alternative, sample length, number and dependence of candidate strategies, Sharpe-ratio sampling law, significance level, retained representation,

and decision rule. An unconditional claim of “too many” Type II errors is therefore incomplete without an alternative and experimental design.

Let C_{FP} be the cost of deploying a strategy without skill and C_{FN} the cost of missing a skilled strategy. The corresponding loss contributions are proportional to $C_{\text{FN}}\beta(\theta_1)$ under the alternative and $C_{\text{FP}}\mathbb{E}_{\theta_0}[\varphi_\alpha]$ under the null. A full decision analysis additionally requires a distribution over null and alternative states. Conservatism may be rational when false deployments are especially costly; greater power may be preferred when missed opportunities dominate. The choice among DSR implementations is therefore partly a decision problem rather than a calibration question alone.

S7.12 Implications for the Unified Framework

Exact uniformity, level validity, conservatism, and power are distinct properties. Calibration must be assessed under the familywise null, not under a mean-zero mixture that usually contains positive-skill strategies. Familywise, selective, posterior, and false-discovery procedures address different targets. DSR-EO, the location construction, and DSR-LS share the same search-adjusted methodology but differ in representation, operational rule, and resulting performance. Richer representations broaden the available analyses, while DSR-EO supplies the benchmark for quantifying the approximation error of tractable procedures such as DSR-LS.

Theorem 1 shows that, in an independent-search benchmark whose selected maximum belongs to a finite-variance GEV domain and satisfies the stated moment-regularity condition, retaining the finite- K location and scale and matching the standardized reference to the extreme-value shape yields asymptotic calibration. Corollary 1 shows that this conclusion is preserved under plug-in estimation when the location and scale errors are negligible relative to σ_K and the estimated extreme-value shape is consistent. These are asymptotic statements and do not replace the exact finite- K DSR-EO benchmark.

S8 Further Extensions of the DSR Methodology

The DSR methodology can continue to be extended without changing its inferential target. Further implementations may enrich the representation of the search-adjusted null distribution F_{M_K} , modify the operational construction applied to that representation, or do both. Null-based quantities are functionals of F_{M_K} ; power also requires a model under the alternative, while posterior false-discovery measures additionally require assumptions about the prevalence and distribution of genuine skill. The inferential scope may therefore expand while the underlying search-adjusted hypothesis remains fixed.

S8.1 Statistical Power

Statistical power is the probability that a specified DSR procedure rejects the search-adjusted null under a specified alternative. Let

$$\text{SR}_A > 0$$

denote the true Sharpe ratio under the alternative, and let $\widehat{\text{SR}}$ be its sample estimate. For a probability-valued DSR procedure conducted at significance level α , power is

$$\pi(\text{SR}_A) = \mathbb{P}_{\text{SR}_A} \left(\text{DSR}(\widehat{\text{SR}}) \geq 1 - \alpha \right).$$

Equivalently, when the procedure is expressed through a search-adjusted p -value, power is

$$\pi(\text{SR}_A) = \mathbb{P}_{\text{SR}_A}(p_{\text{DSR}} \leq \alpha).$$

These formulations describe the same rejection rule when

$$p_{\text{DSR}} = 1 - \text{DSR}.$$

Power is a property of the complete procedure under the specified alternative. It may depend on SR_A , the sample length, the number and dependence of the candidate strategies, the sampling law of the selected Sharpe ratio, the representation retained from F_{M_K} , and the operational decision rule. Neither μ_K nor (μ_K, σ_K) alone determines power. Under a maintained location-and-scale approximation, these moments provide null-distribution inputs, while the alternative model and rejection rule complete the calculation.

S8.2 Search-Adjusted False Discovery Rates

Familywise significance and posterior false-discovery probability answer different questions. The former controls erroneous rejection under the global null; the latter asks whether a reported discovery reflects genuine skill. Planned and observed false-discovery rates are therefore natural extensions of search-adjusted inference. López de Prado and Fabozzi (2026) show that, with latent specification search, the posterior false-discovery probability is generally not identifiable from in-sample evidence without modelling the search process.

Such measures require the prevalence and distribution of genuinely profitable strategies, the rejection probability under the search-adjusted null, the power of the chosen procedure, and the rule defining a discovery. A richer representation of F_{M_K} may supply the required null inputs, but it cannot replace assumptions about the alternative or the prevalence of skill. False-discovery analysis therefore continues and extends the DSR methodology without redefining its familywise inferential target.

S8.3 Exact Finite-Sample Inference

Exact finite-sample inference is not a new construction of the present article. López de Prado et al. (2026, equations 26–27) derive the finite- K order-statistic distribution and its corresponding FWER-controlling threshold under the maintained independent Gaussian search model. When dependence is summarized through an effective number of independent trials, the same order-statistic formula is an operational approximation whose calibration depends on the adequacy of that reduction. For taxonomic convenience, the present article labels the independent-Gaussian construction DSR-EO. More generally, whenever the complete null distribution F_{M_K} is available, an exact implementation may use the critical value $c_{\alpha, K}^{\text{ex}} = F_{M_K}^{-1}(1 - \alpha)$ from equation (13) or the upper-tail p -value $p_K^{\text{ex}} = 1 - F_{M_K}(M_K)$ from equation (14), subject to the stated continuity and null-distribution conditions.

S8.4 Future Implementations

The continuing DSR methodology imposes no restriction on the representation or computational method used to implement search-adjusted inference. Future procedures may employ:

- higher-order moment approximations;
- parametric tail approximations;
- saddlepoint approximations;
- Edgeworth expansions;
- bootstrap distributions or bootstrap critical values;
- Monte Carlo approximations of F_{M_K} , its quantiles, or its p -values;
- exact finite-sample distributions;
- numerical integration under specified dependence models; or
- representations and operational constructions not yet developed.

For comparative analysis, a complete description of an implementation records the retained representation, the critical-value, p -value, or decision-rule construction, and the assumptions under which calibration and power are evaluated. The earlier DSR papers already contain these elements; the present framework makes them explicit in a common notation. Improvements in computation, approximation, or inferential scope refine the implementation rather than redefine the methodology. The common foundation remains inference relative to the null distribution generated by the strategy-selection process.

A Proofs for the Representation Results

Proof of Proposition 1. By Definition 1, every DSR implementation evaluates an observed Sharpe ratio relative to F_{M_K} . Operationally, it retains a representation $\mathcal{R}(F_{M_K})$ and applies a construction Ψ to that representation. Implementations can therefore be classified by the information they retain from the common null distribution. The representation alone does not completely determine the implementation, because the operational construction must also be specified. $\square$

Proof of Proposition 2. Let $\mathcal{R}_L(F_{M_K}) = \mu_K$, as defined in equation (4), denote the location representation, and let $\mathcal{R}_{LS}(F_{M_K}) = (\mu_K, \sigma_K)$, as defined in equation (6), denote the corresponding location-and-scale representation. Define the projection onto the first component by

$$\text{pr}_1(\mu_K, \sigma_K) = \mu_K.$$

Then

$$\mathcal{R}_L(F_{M_K}) = \text{pr}_1(\mathcal{R}_{LS}(F_{M_K})).$$

Consequently, every inferential quantity that depends only on μ_K can also be computed from $\mathcal{R}_{LS}(F_{M_K})$ in equation (6). The converse does not generally hold, because quantities that depend on σ_K , defined through equation (5), cannot be recovered from μ_K alone. The location representation is therefore informationally nested within the location-and-scale representation. $\square$

B Technical Proofs for the Calibration Results

This appendix provides the proofs underlying the operational-calibration results in Section S7. The first subsection proves the general calibration-distortion identity. The remaining subsections concern the independent-Gaussian benchmark.

B.1 Proof of the Calibration-Distortion Identity

Proof of Proposition 3. By the definition of F_{M_K} ,

$$\mathbb{P}_0(M_K > c_{\alpha,K}^G) = 1 - F_{M_K}(c_{\alpha,K}^G).$$

Since G_K is continuous and strictly increasing, its inverse is well defined on $(0, 1)$ and

$$G_K(c_{\alpha,K}^G) = G_K(G_K^{-1}(1 - \alpha)) = 1 - \alpha.$$

Therefore,

$$\begin{aligned} \Delta_{\alpha,K}^G &= 1 - F_{M_K}(c_{\alpha,K}^G) - \alpha \\ &= G_K(c_{\alpha,K}^G) - F_{M_K}(c_{\alpha,K}^G), \end{aligned}$$

which proves equation (11).

Continuity and strict monotonicity of F_{M_K} imply

$$F_{M_K}(c_{\alpha,K}^{\text{ex}}) = F_{M_K}(F_{M_K}^{-1}(1 - \alpha)) = 1 - \alpha.$$

It follows immediately that the approximate rule is conservative, exactly calibrated, or liberal according as $c_{\alpha,K}^G$ is greater than, equal to, or smaller than $c_{\alpha,K}^{\text{ex}}$.

It remains to prove the uniform identity. For $u \in (0, 1)$, set

$$x_u := G_K^{-1}(1 - u).$$

Because G_K is strictly increasing,

$$\begin{aligned} \mathbb{P}_0(p_K^G \leq u) &= \mathbb{P}_0(1 - G_K(M_K) \leq u) \\ &= \mathbb{P}_0(G_K(M_K) \geq 1 - u) \\ &= \mathbb{P}_0(M_K \geq x_u). \end{aligned}$$

In general,

$$\mathbb{P}_0(M_K \geq x_u) = 1 - F_{M_K}(x_u^-),$$

where $F_{M_K}(x^-)$ denotes the left limit at x . Since F_{M_K} is continuous, $F_{M_K}(x_u^-) = F_{M_K}(x_u)$, and therefore

$$\mathbb{P}_0(p_K^G \leq u) = 1 - F_{M_K}(x_u).$$

Also, $G_K(x_u) = 1 - u$, so

$$\begin{aligned}\mathbb{P}_0(p_K^G \leq u) - u &= 1 - F_{M_K}(x_u) - u \\ &= G_K(x_u) - F_{M_K}(x_u).\end{aligned}$$

Finally, a continuous and strictly increasing CDF maps $\mathbb{R}$ bijectively onto $(0, 1)$. Thus, as u ranges over $(0, 1)$, the point $x_u = G_K^{-1}(1 - u)$ ranges over all of $\mathbb{R}$. Taking absolute values and suprema gives equation (12). $\square$

For the independent-Gaussian benchmark, let $Z_1, \dots, Z_K$ and M_K be as specified in equations (17) and (18), respectively, and let Φ and ϕ denote the standard Normal distribution function and density. Define μ_K , σ_K^2 , and $z_{1-\alpha}$ as in equations (4), (5), and (20), respectively.

B.2 Exact Distribution, Moments, and Calibration

Since the variables $Z_1, \dots, Z_K$ are independent,

$$\begin{aligned}F_{M_K}(x) &= \mathbb{P}(M_K \leq x) \\ &= \mathbb{P}(Z_1 \leq x, \dots, Z_K \leq x) \\ &= \prod_{i=1}^K \mathbb{P}(Z_i \leq x) \\ &= \Phi(x)^K.\end{aligned}$$

Consequently, the density of M_K is

$$f_{M_K}(x) = K\phi(x)\Phi(x)^{K-1}.$$

It follows that μ_K and $\mathbb{E}[M_K^2]$ are given by equations (23) and (24), respectively, with σ_K^2 also specified in equation (24).

The exact upper-tail critical value at significance level α is the solution of

$$\mathbb{P}(M_K > c_{\alpha, K}^{\text{ex}}) = \alpha.$$

Equivalently,

$$\Phi(c_{\alpha, K}^{\text{ex}})^K = 1 - \alpha,$$

and hence yields the closed form in equation (21).

The corresponding exact upper-tail p -value is p_K^{ex} in equation (22).

Because F_{M_K} is continuous, the probability integral transform gives

$$F_{M_K}(M_K) \sim \mathcal{U}(0, 1).$$

Since $1 - U \sim \mathcal{U}(0, 1)$ whenever $U \sim \mathcal{U}(0, 1)$, it follows that

$$p_K^{\text{ex}} \sim \mathcal{U}(0, 1).$$

Therefore,

$$\mathbb{P}(p_K^{\text{ex}} \leq \alpha) = \alpha,$$

and the exact order-statistic rule has finite-sample rejection probability exactly equal to its nominal level under the maintained simple null.

B.3 Proof That Moments Do Not Determine Exact Critical Values

Proof of Proposition 4. Let $m_r(F)$ denote the vector containing the first r moments of a distribution F . By assumption,

$$m_r(F_1) = m_r(F_2),$$

but

$$F_1^{-1}(1 - \alpha) \neq F_2^{-1}(1 - \alpha).$$

Suppose that a critical-value rule depends only on the first r moments. Then it can be written as

$$c_\alpha(F) = h_\alpha(m_r(F))$$

for some function h_α . Since F_1 and F_2 have identical first r moments,

$$c_\alpha(F_1) = h_\alpha(m_r(F_1)) = h_\alpha(m_r(F_2)) = c_\alpha(F_2).$$

Thus, the rule assigns a common critical value c_α to both distributions.

Because F_1 and F_2 are continuous and strictly increasing, exact upper-tail rejection probability α under F_j requires

$$1 - F_j(c_\alpha) = \alpha,$$

or equivalently,

$$c_\alpha = F_j^{-1}(1 - \alpha).$$

Exact calibration under both distributions would therefore require

$$c_\alpha = F_1^{-1}(1 - \alpha) = F_2^{-1}(1 - \alpha),$$

contradicting the assumption that the two $(1 - \alpha)$ -quantiles differ. Hence no critical-value rule depending only on the first r moments can have rejection probability exactly equal to α under both F_1 and F_2 . $\square$

The result concerns exact upper-tail calibration. It does not exclude the construction of conservative procedures from moment inequalities under additional assumptions. It establishes that a finite collection of moments does not, in general, identify the exact upper-tail quantile without a further distributional or approximation assumption.

B.4 Gaussian Extreme-Value Normalization

For $K \geq 3$, define

$$b_K := \Phi^{-1}\left(1 - \frac{1}{K}\right), \quad a_K := \frac{1}{b_K}.$$

By construction,

$$K\bar{\Phi}(b_K) = 1, \quad \bar{\Phi}(x) := 1 - \Phi(x).$$

For every fixed $x \in \mathbb{R}$, Mills' ratio gives

$$\frac{\bar{\Phi}(b_K + a_K x)}{\bar{\Phi}(b_K)} \longrightarrow e^{-x}.$$

Indeed,

$$\begin{aligned} \frac{\bar{\Phi}(b_K + x/b_K)}{\bar{\Phi}(b_K)} &\sim \frac{b_K}{b_K + x/b_K} \exp\left[-\frac{1}{2} \left\{ \left(b_K + \frac{x}{b_K}\right)^2 - b_K^2 \right\}\right] \\ &= \frac{b_K^2}{b_K^2 + x} \exp\left(-x - \frac{x^2}{2b_K^2}\right) \\ &\longrightarrow e^{-x}. \end{aligned}$$

Therefore,

$$K\bar{\Phi}(b_K + a_K x) \longrightarrow e^{-x}.$$

It follows that

$$\begin{aligned} \mathbb{P}\left(\frac{M_K - b_K}{a_K} \leq x\right) &= \Phi(b_K + a_K x)^K \\ &= \left[1 - \bar{\Phi}(b_K + a_K x)\right]^K \\ &\longrightarrow \exp(-e^{-x}). \end{aligned}$$

Hence,

$$\frac{M_K - b_K}{a_K} \xrightarrow{d} G, \tag{96}$$

where G is a standard Gumbel random variable with the distribution given in equation (28).

We now verify the moment-convergence step. Set

$$Y_K := \frac{M_K - b_K}{a_K} = b_K(M_K - b_K).$$

It is enough to prove that $\{Y_K^2 : K \geq 3\}$ is uniformly integrable. The finitely many indices for which $b_K < 2$ are immaterial, so assume $b_K \geq 2$. The standard Mills inequalities

$$\frac{x}{1+x^2}\phi(x) \leq \bar{\Phi}(x) \leq \frac{\phi(x)}{x}, \quad x > 0,$$

imply, for every $y \geq 0$,

$$\begin{aligned}\mathbb{P}(Y_K > y) &\leq K\bar{\Phi}\left(b_K + \frac{y}{b_K}\right) \\ &= \frac{\bar{\Phi}(b_K + y/b_K)}{\bar{\Phi}(b_K)} \\ &\leq 2e^{-y}.\end{aligned}$$

For the lower tail, if $0 \leq y \leq b_K^2/2$, then $t = b_K - y/b_K \geq b_K/2$, and the same inequalities give

$$K\bar{\Phi}(t) = \frac{\bar{\Phi}(t)}{\bar{\Phi}(b_K)} \geq \frac{1}{2} \exp\left(\frac{b_K^2 - t^2}{2}\right) \geq \frac{1}{2} e^{3y/4}.$$

Therefore,

$$\mathbb{P}(Y_K < -y) = \Phi(t)^K \leq \exp\left(-\frac{1}{2}e^{3y/4}\right), \quad 0 \leq y \leq b_K^2/2.$$

If $b_K^2/2 \leq y \leq b_K^2$, monotonicity and Mills' inequalities yield

$$\mathbb{P}(Y_K < -y) \leq \Phi(b_K/2)^K \leq \exp(-e^{3b_K^2/8}) \leq \exp(-e^{3y/8}).$$

Finally, for $y \geq b_K^2$, write $u = (y - b_K^2)/b_K \geq 0$. The Gaussian Chernoff bound gives

$$\mathbb{P}(Y_K < -y) = \bar{\Phi}(u)^K \leq 2^{-K} \exp\left(-\frac{Ku^2}{2}\right).$$

The first two bounds are integrable against y , uniformly in K . Moreover, the same Chernoff bound and $\bar{\Phi}(b_K) = 1/K$ imply $b_K^2 = O(\log K)$. For the last region, the change of variables $y = b_K^2 + b_K u$ shows that its contribution to the second tail moment is bounded by

$$2^{-K} \left[O(b_K^4) + O\left(\frac{b_K^3}{\sqrt{K}}\right) + O\left(\frac{b_K^2}{K}\right) \right],$$

which converges to zero. The tail-integral criterion now proves uniform integrability of Y_K^2 . Consequently, convergence in distribution is accompanied by convergence of the first two moments:

$$\mathbb{E}\left[\frac{M_K - b_K}{a_K}\right] \longrightarrow \mathbb{E}[G] = \gamma,$$

and

$$\text{Var}\left(\frac{M_K - b_K}{a_K}\right) \longrightarrow \text{Var}(G) = \frac{\pi^2}{6},$$

where γ denotes Euler's constant. Therefore,

$$\mu_K = b_K + \gamma a_K + o(a_K) \tag{97}$$

and

$$\sigma_K = \frac{\pi}{\sqrt{6}} a_K + o(a_K). \tag{98}$$

B.5 Asymptotic Conservatism of the Location Analogue

Proof of the first assertion of Proposition 5. Fix $\alpha \in (0, 1/2)$, so that

$$z_{1-\alpha} > 0.$$

From the Gaussian extreme-value moment expansion,

$$\mu_K = b_K + O(b_K^{-1}),$$

and therefore

$$\mu_K - b_K \longrightarrow 0.$$

Consequently, for all sufficiently large K ,

$$\mu_K + z_{1-\alpha} \geq b_K + \frac{z_{1-\alpha}}{2}.$$

Since $\bar{\Phi}$ is decreasing,

$$K\bar{\Phi}(\mu_K + z_{1-\alpha}) \leq K\bar{\Phi}\left(b_K + \frac{z_{1-\alpha}}{2}\right).$$

Using $K\bar{\Phi}(b_K) = 1$, the right-hand side becomes

$$\frac{\bar{\Phi}(b_K + z_{1-\alpha}/2)}{\bar{\Phi}(b_K)}.$$

Mills' ratio yields

$$\begin{aligned} \frac{\bar{\Phi}(b_K + z_{1-\alpha}/2)}{\bar{\Phi}(b_K)} &\sim \frac{b_K}{b_K + z_{1-\alpha}/2} \\ &\quad \times \exp\left[-\frac{1}{2}\left\{\left(b_K + \frac{z_{1-\alpha}}{2}\right)^2 - b_K^2\right\}\right] \\ &= \frac{b_K}{b_K + z_{1-\alpha}/2} \exp\left[-\frac{z_{1-\alpha}}{2}b_K - \frac{z_{1-\alpha}^2}{8}\right] \\ &\longrightarrow 0. \end{aligned}$$

Thus,

$$K\bar{\Phi}(\mu_K + z_{1-\alpha}) \longrightarrow 0.$$

Let

$$x_K := \bar{\Phi}(\mu_K + z_{1-\alpha}).$$

Then $Kx_K \rightarrow 0$, and therefore

$$\begin{aligned} \Phi(\mu_K + z_{1-\alpha})^K &= (1 - x_K)^K \\ &= \exp\{K \log(1 - x_K)\} \\ &\longrightarrow 1. \end{aligned}$$

It follows that the convergence asserted in equation (27) holds.

Hence the location analogue becomes asymptotically conservative, with its rejection probability converging to zero. $\square$

B.6 Limiting Calibration of the Gaussian Location-and-Scale Analogue

Proof of the second assertion of Proposition 5 . Write

$$Y_K := \frac{M_K - b_K}{a_K}.$$

The preceding extreme-value argument gives

$$Y_K \xrightarrow{d} G.$$

Moreover, the expansions for μ_K and σ_K are given in equations (97) and (98), respectively. Therefore,

$$\begin{aligned} \frac{M_K - \mu_K}{\sigma_K} &= \frac{M_K - b_K - \gamma a_K + o(a_K)}{(\pi/\sqrt{6})a_K + o(a_K)} \\ &= \frac{Y_K - \gamma + o(1)}{\pi/\sqrt{6} + o(1)}. \end{aligned}$$

By Slutsky's theorem, this yields the convergence result in equation (29).

The limiting distribution is continuous. Consequently,

$$\begin{aligned} \lim_{K \rightarrow \infty} \mathbb{P}(M_K > \mu_K + \sigma_K z_{1-\alpha}) \\ = \mathbb{P}\left(\frac{G - \gamma}{\pi/\sqrt{6}} > z_{1-\alpha}\right). \end{aligned}$$

The event on the right is equivalent to

$$G > \gamma + \frac{\pi}{\sqrt{6}} z_{1-\alpha}.$$

Using the distribution in equation (28), we obtain

$$\begin{aligned} \mathbb{P}\left(G > \gamma + \frac{\pi}{\sqrt{6}} z_{1-\alpha}\right) \\ = 1 - \exp\left[-\exp\left\{-\gamma - \frac{\pi}{\sqrt{6}} z_{1-\alpha}\right\}\right]. \end{aligned}$$

Hence, the limit is the one stated in equation (30).

At nominal significance level $\alpha = 0.05$,

$$z_{1-\alpha} = z_{0.95},$$

and the limiting rejection probability is

$$1 - \exp\left[-\exp\left\{-\gamma - \frac{\pi}{\sqrt{6}} z_{0.95}\right\}\right] \approx 0.0658301.$$

Thus, at a nominal 5% level, the Gaussian location-and-scale analogue has a limiting rejection probability of approximately 6.583%, or a positive calibration gap of approximately 1.583 percentage points. This gap quantifies the asymptotic effect of using a Gaussian moment-matched approximation in place of the standardized Gumbel limit of the selected maximum. $\square$

B.7 Scope of the Gaussian Benchmark

The preceding results concern the transparent standardized rules $M_K > \mu_K + z_{1-\alpha}$ and $M_K > \mu_K + \sigma_K z_{1-\alpha}$, associated with equations (25) and (26), respectively.

They provide a transparent calibration benchmark for quantifying how location and location-and-scale approximations behave as K grows. This is fully consistent with the 2026 distinction between the practical DSR-LS approximation and the calibrated finite- K DSR-EO solution.

Mapping the standardized analogue to every component of the operational 2014 and 2026 formulas requires a common finite-sample derivation that distinguishes:

1. the sampling uncertainty of the estimated Sharpe ratio;
2. the cross-strategy dispersion induced by the research process;
3. the scale entering the Probabilistic Sharpe Ratio;
4. finite-sample skewness and kurtosis corrections;
5. the dependence structure among candidate strategies; and
6. the distributional approximation imposed on the selected maximum.

The Gaussian calculations should therefore be interpreted as an analytical benchmark for the moment-matching component of DSR-LS. The resulting bridge will determine when sampling uncertainty and the skewness and kurtosis corrections close, reverse, or leave unchanged the calibration gap relative to DSR-EO.

C Technical Proofs for Student- t Calibration

C.1 Exact Finite- K Calibration

Proof of Proposition 6. By (35),

$$\begin{aligned}\mathbb{P}(M_{K,\nu} > c_{\alpha,K,\nu}^{LS}) &= 1 - F_{K,\nu}(c_{\alpha,K,\nu}^{LS}) \\ &= 1 - T_\nu(c_{\alpha,K,\nu}^{LS})^K.\end{aligned}$$

Substitution of (39) proves (40). Furthermore,

$$\begin{aligned}\pi_{\alpha,K,\nu}^{LS} \leq \alpha &\iff T_\nu(c_{\alpha,K,\nu}^{LS})^K \geq 1 - \alpha \\ &\iff T_\nu(c_{\alpha,K,\nu}^{LS}) \geq (1 - \alpha)^{1/K} \\ &\iff c_{\alpha,K,\nu}^{LS} \geq T_\nu^{-1}\left((1 - \alpha)^{1/K}\right),\end{aligned}$$

where the last equivalence follows from the strict monotonicity of T_ν . This proves (41) and the stated classification. $\square$

C.2 Moment Existence and Scale Invariance

For every $r > 0$, the upper tail of $M_{K,\nu}$ has the same tail index ν as the individual Student variable. Hence

$$\mathbb{E}[(M_{K,\nu})_+^r] < \infty \iff r < \nu. \quad (99)$$

In particular, $M_{K,\nu}$ has a finite absolute first moment exactly when $\nu > 1$, and a finite second moment exactly when $\nu > 2$. The first-two-moment location-and-scale rule is therefore not defined for $\nu \leq 2$.

The calibration results are invariant to positive rescaling. For example, replacing X_i by

$$\tilde{X}_i = \sqrt{\frac{\nu-2}{\nu}} X_i, \quad \nu > 2, \quad (100)$$

standardizes the marginal variance to one but multiplies $\mu_{K,\nu}$, $\sigma_{K,\nu}$, and each location-and-scale or exact critical value by the same positive factor. The corresponding rejection probabilities are unchanged.

C.3 A Tail-Stable Representation for Numerical Evaluation

Let U_K have the $\text{Beta}(K, 1)$ distribution. By the probability integral transform and standard order-statistic theory (David and Nagaraja 2003, Chapter 2),

$$M_{K,\nu} \stackrel{d}{=} T_\nu^{-1}(U_K). \quad (101)$$

If $W \sim \text{Exp}(1)$ and $U_K = \exp(-W/K)$, then $\mathbb{P}(U_K \leq u) = u^K$. Consequently, for $r \in \{1, 2\}$ (whenever the corresponding moment is finite),

$$\mathbb{E}[M_{K,\nu}^r] = \int_0^\infty [T_\nu^{-1}(e^{-w/K})]^r e^{-w} dw. \quad (102)$$

Equation (102) is numerically preferable to direct integration over the far upper tail. In implementation, the survival probability should be evaluated as $1 - e^{-w/K} = -\text{expm1}(-w/K)$ to avoid cancellation near $w = 0$.

As an independent check, the same moments can be evaluated without quantile inversion. Whenever the relevant moments are finite,

$$\mathbb{E}[M_{K,\nu}] = \int_0^\infty \{1 - T_\nu(x)^K\} dx - \int_0^\infty T_\nu(-x)^K dx, \quad (103)$$

$$\mathbb{E}[M_{K,\nu}^2] = 2 \int_0^\infty x [1 - T_\nu(x)^K + T_\nu(-x)^K] dx. \quad (104)$$

These identities follow from the standard tail-integral formulas for the first and second moments. Numerical evaluation of (103)–(104) reproduces the moments underlying both reference-shape calculations in Supplementary Exhibit 2 independently of (102).

C.4 Large- K Calibration

The Student- t_ν survival function is regularly varying:

$$1 - T_\nu(x) \sim C_\nu x^{-\nu}, \quad x \rightarrow \infty, \quad (105)$$

where

$$C_\nu = \frac{\Gamma((\nu+1)/2) \nu^{\nu/2-1}}{\sqrt{\pi} \Gamma(\nu/2)}. \quad (106)$$

To verify (105)–(106), write

$$t_\nu(x) = A_\nu \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}, \quad A_\nu = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi} \Gamma(\nu/2)}.$$

Then

$$t_\nu(x) \sim A_\nu \nu^{(\nu+1)/2} x^{-(\nu+1)}.$$

An application of l'Hôpital's rule to $\bar{T}_\nu(x)/x^{-\nu}$ yields

$$\bar{T}_\nu(x) \sim \frac{A_\nu \nu^{(\nu+1)/2}}{\nu} x^{-\nu} = C_\nu x^{-\nu},$$

which gives the stated constant.

Proof of Proposition 7. Write $\bar{T}_\nu = 1 - T_\nu$. Regular variation and the definition of $a_{K,\nu}$ imply, for every $y > 0$,

$$K\bar{T}_\nu(a_{K,\nu}y) \longrightarrow y^{-\nu}.$$

Hence

$$\begin{aligned} \mathbb{P}\left(\frac{M_{K,\nu}}{a_{K,\nu}} \leq y\right) &= T_\nu(a_{K,\nu}y)^K \\ &= \left[1 - \bar{T}_\nu(a_{K,\nu}y)\right]^K \\ &\longrightarrow \exp(-y^{-\nu}), \end{aligned}$$

which proves (43).

We next establish moment convergence. Fix $r < \nu$ and choose $\varepsilon \in (0, \nu - r)$. Potter bounds for regularly varying functions (Bingham, Goldie, and Teugels 1987, Theorem 1.5.6) imply that, for all sufficiently large K and every $y \geq 1$,

$$K\bar{T}_\nu(a_{K,\nu}y) \leq Cy^{-\nu+\varepsilon}$$

for a finite constant C . Since $1 - (1 - p)^K \leq Kp$,

$$\mathbb{P}\left(\frac{M_{K,\nu}}{a_{K,\nu}} > y\right) \leq Cy^{-\nu+\varepsilon}, \quad y \geq 1.$$

To make the uniform-integrability step explicit, let $Z_{K,\nu} = (M_{K,\nu}/a_{K,\nu})_+$. For $A \geq 1$,

$$\begin{aligned} \mathbb{E}\left[Z_{K,\nu}^r \mathbf{1}_{\{Z_{K,\nu} > A\}}\right] &= A^r \mathbb{P}(Z_{K,\nu} > A) + \int_A^\infty ry^{r-1} \mathbb{P}(Z_{K,\nu} > y) dy \\ &\leq CA^{r-\nu+\varepsilon} + Cr \int_A^\infty y^{r-1-\nu+\varepsilon} dy. \end{aligned}$$

Because $r < \nu - \varepsilon$, the right-hand side converges to zero as $A \rightarrow \infty$, uniformly in all sufficiently large K . Thus the positive parts are uniformly integrable through every order $r < \nu$.

The negative part is negligible. On the event $\{M_{K,\nu} < 0\}$, all X_j are negative and $(-M_{K,\nu})^r \leq (-X_1)^r$. Hence, by independence and symmetry,

$$\begin{aligned} \mathbb{E}[(-M_{K,\nu})_+^r] &\leq \mathbb{E}\left[(-X_1)_+^r \prod_{j=2}^K \mathbf{1}_{\{X_j < 0\}}\right] \\ &= \mathbb{E}[(-X_1)_+^r] 2^{-(K-1)}. \end{aligned}$$

After division by $a_{K,\nu}^r$, this converges to zero. Therefore the first and second moments converge because $\nu > 2$. For a Fréchet variable (Kotz and Nadarajah 2000, Chapter 1),

$$\mathbb{E}[Y_\nu^r] = \Gamma\left(1 - \frac{r}{\nu}\right), \quad r < \nu.$$

Taking $r = 1$ and $r = 2$ yields convergence of the first and second raw moments, and hence (44) and (45).

Finally,

$$\frac{c_{\alpha,K,\nu}^{LS}}{a_{K,\nu}} = \frac{\mu_{K,\nu}}{a_{K,\nu}} + \frac{\sigma_{K,\nu}}{a_{K,\nu}} z_{1-\alpha} \longrightarrow m_\nu + s_\nu z_{1-\alpha}.$$

Set

$$d_{\alpha,\nu} := m_\nu + s_\nu z_{1-\alpha}.$$

Because $\alpha < 1/2$, we have $z_{1-\alpha} > 0$, and therefore $d_{\alpha,\nu} > 0$. By (43) and the preceding convergence of the normalized critical value,

$$\begin{aligned} \mathbb{P}\left(M_{K,\nu} \leq c_{\alpha,K,\nu}^{LS}\right) &= \mathbb{P}\left(\frac{M_{K,\nu}}{a_{K,\nu}} \leq \frac{c_{\alpha,K,\nu}^{LS}}{a_{K,\nu}}\right) \\ &\longrightarrow \mathbb{P}(Y_\nu \leq d_{\alpha,\nu}) \\ &= \exp\left(-d_{\alpha,\nu}^{-\nu}\right), \end{aligned}$$

because the Fréchet distribution function is continuous at the positive point $d_{\alpha,\nu}$. Taking complements yields

$$\mathbb{P}\left(M_{K,\nu} > c_{\alpha,K,\nu}^{LS}\right) \longrightarrow 1 - \exp\left[-(m_\nu + s_\nu z_{1-\alpha})^{-\nu}\right],$$

which is (46). □

C.5 Asymptotic Calibration of the Standardized Fréchet Reference

Proof of the Fréchet-reference calibration corollary. The finite- K expression follows immediately from the exact maximum CDF in (35) and the definition (50):

$$\mathbb{P}\left(M_{K,\nu} > c_{\alpha,K,\nu}^{FLS}\right) = 1 - T_\nu\left(c_{\alpha,K,\nu}^{FLS}\right)^K,$$

which is (51).

For asymptotic calibration, Proposition 7 gives

$$\frac{M_{K,\nu}}{a_{K,\nu}} \xrightarrow{d} Y_\nu, \quad \frac{\mu_{K,\nu}}{a_{K,\nu}} \rightarrow m_\nu, \quad \frac{\sigma_{K,\nu}}{a_{K,\nu}} \rightarrow s_\nu.$$

Consequently,

$$\frac{M_{K,\nu} - \mu_{K,\nu}}{\sigma_{K,\nu}} \xrightarrow{d} \frac{Y_\nu - m_\nu}{s_\nu}.$$

The limiting standardized Fréchet law is continuous, and by (49) its $(1 - \alpha)$ -quantile is $q_{1-\alpha}^{F_\nu}$. Therefore

$$\begin{aligned} \mathbb{P}\left(M_{K,\nu} > c_{\alpha,K,\nu}^{FLS}\right) &= \mathbb{P}\left(\frac{M_{K,\nu} - \mu_{K,\nu}}{\sigma_{K,\nu}} > q_{1-\alpha}^{F_\nu}\right) \\ &\longrightarrow \alpha, \end{aligned}$$

which proves (52). □

The $(1 - \alpha)$ -quantile of Y_ν is

$$q_{\nu, 1-\alpha} = [-\log(1 - \alpha)]^{-1/\nu}.$$

The limiting rule is non-liberal precisely when its limiting critical value $m_\nu + s_\nu z_{1-\alpha}$ is at least this exact Fréchet quantile. Substitution of m_ν and s_ν gives (47).

C.6 Numerical Implications at the Five-Percent Level

At $\alpha = 0.05$, numerical root finding for

$$1 - \exp\left\{- (m_\nu + s_\nu z_{0.95})^{-\nu}\right\} = 0.05 \quad (107)$$

returns

$$\nu_\infty^* \approx 3.73871. \quad (108)$$

On a dense numerical grid over $2.001 \leq \nu \leq 20$, the limiting-size function exhibits a single crossing: it is below 0.05 to the left of this value and above 0.05 to the right. This sign classification is numerical and restricted to the examined range; no analytical monotonicity or uniqueness theorem in ν is asserted here.

The corresponding finite- K roots located numerically on the same degrees-of-freedom range are reported in Supplementary Exhibit 3. As above, the supplementary exhibit records numerical roots and does not assert analytical uniqueness.

Supplementary Exhibit 3: Numerical degrees-of-freedom crossing at $\alpha = 0.05$

K	ν_K^*
10	3.5322
100	3.6668
1000	3.7164
∞	3.7387

A practical accuracy criterion can also be read numerically from the limiting size. At nominal level 5%, numerical solution of the relevant level-set equations on $2.001 \leq \nu \leq 20$ indicates that the connected accuracy region detected on this range lies within 0.5 percentage points of nominal for approximately

$$3.241 \leq \nu \leq 4.585, \quad (109)$$

and within one percentage point on the same examined range for approximately

$$2.908 \leq \nu \leq 6.472. \quad (110)$$

Let G have the standard Gumbel distribution $\mathbb{P}(G \leq x) = \exp(-e^{-x})$. The Fréchet variable satisfies $Y_\nu \stackrel{d}{=} \exp(G/\nu)$. Consequently,

$$m_\nu = 1 + \frac{\gamma}{\nu} + O(\nu^{-2}), \quad s_\nu = \frac{\pi}{\sqrt{6}\nu} + O(\nu^{-2}),$$

where γ is Euler's constant. It follows that, for fixed $\alpha \in (0, 1/2)$,

$$\pi_{\alpha, \nu}^{LS, \infty} \longrightarrow 1 - \exp\left\{-\exp\left[-\gamma - \frac{\pi}{\sqrt{6}} z_{1-\alpha}\right]\right\} \quad (\nu \rightarrow \infty).$$

At $\alpha = 0.05$, this limit is approximately 0.0658301, the standardized-Gumbel benchmark for independent Gaussian experiments. This is the iterated limit obtained by first taking $K \rightarrow \infty$ at fixed ν and then letting $\nu \rightarrow \infty$; no uniformity in the two limits is asserted.

References

- Bailey, David H., and Marcos López de Prado. 2012. “The Sharpe Ratio Efficient Frontier.” *Journal of Risk* 15 (2): 3–44.
- . 2014. “The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality.” *Journal of Portfolio Management* 40 (5): 94–107.
- Bingham, Nicholas H., Charles M. Goldie, and Jef L. Teugels. 1987. *Regular Variation*. Vol. 27. Encyclopedia of Mathematics and Its Applications. Cambridge: Cambridge University Press.
- David, Herbert A., and Haikady N. Nagaraja. 2003. *Order Statistics*. 3rd ed. Hoboken, NJ: John Wiley & Sons.
- Kotz, Samuel, and Saralees Nadarajah. 2000. *Extreme Value Distributions: Theory and Applications*. London: Imperial College Press.
- López de Prado, Marcos, and Frank J. Fabozzi. 2026. “The False Discovery Rate in Finance: Identification Failure and Search-Adjusted Estimation.” Working paper, SSRN 6450418.
- López de Prado, Marcos, Alexander Lipton, and Vincent Zoonekynd. 2026. “Sharpe Ratio Inference: A New Standard for Decision Making and Reporting.” *Journal of Portfolio Management* 52 (6): 50–66.